Paving the Way Toward Foundation Models for Irregular and Unaligned Satellite Image Time Series

Iris Dumeur, Silvia Valero, and Jordi Inglada

***Abstract*— Although recently several foundation models (FMs) for satellite remote sensing (RS) imagery have been proposed, they fail to address major challenges of operational applications. Indeed, representations that do not take into account the spectral, spatial, and temporal dimensions of the data as well as the irregular or unaligned temporal sampling are of little use for most real-world applications. As a consequence, we address some existing shortcomings in the design of FMs for RS data. In particular, we propose an ALIgned Sits Encoder (ALISE), a novel approach that leverages the spatial, spectral, and temporal dimensions of irregular and unaligned satellite image time series (SITS) while producing aligned latent representations. ALISE provides easy-to-use fixed-size SITS representations that preserve the spatial resolution of the input SITS required for most mapping tasks. Moreover, to learn informative representations of SITS, we investigate the integration of instance discrimination losses within a masked autoencoding pretraining task, utilizing a multiview framework. The model is pretrained on a custom-built Sentinel-2 (S2) multiyear SITS unlabeled dataset. The genericity of the provided representations is assessed on three downstream tasks: crop segmentation, land cover segmentation, and an unsupervised crop change detection task. The results suggest that the use of ALISE's aligned representations is significantly more effective than previous self-supervised learning (SSL) methods for linear-probing segmentation tasks. Additionally, the experiments show the interest of using ALISE representations for unsupervised change detection. Finally, the impact of the pretraining hyperparameters and the proposed method for aligning irregular and unaligned time series are examined in detail. The code, the pretrained model, and the datasets are released at https://src.koda.cnrs.fr/iris.dumeur/alise**

***Index Terms*— Foundation model (FM), representation learning, self-supervised learning (SSL), satellite image time series (SITS).**

I. Introduction

OVER the past decade, numerous Earth observation satellite missions have been launched to monitor the effects of climate change on land surfaces. In particular, missions combining high spectral and spatial resolutions with high temporal revisit, such as Sentinel-2 (S2) [1], enable an exhaustive and systematic capture of land surfaces. The acquired data correspond to satellite image time series (SITS), which are 4-D objects encompassing spatial, spectral, and temporal dimensions. By nature, SITS serve as essential inputs for generating frequently large-scale land cover segmentation maps required by the climate and geoscience community. Currently, numerous deep learning (DL) studies have been proposed to exploit SITS for producing high-resolution maps characterizing land surfaces. For instance, several works have addressed the production of crop semantic segmentation maps [2], [3], [4]. Nevertheless, a major limitation of supervised DL technologies is their need for large amounts of labeled data. Existing supervised methods are therefore ill-suited to the production of large-scale maps, covering several time periods and/or the entire globe. In addition, DL methods applied to SITS are task-specific and thus not adapted to solving the multiple tasks required for Earth monitoring. Therefore, we propose to investigate the production of SITS representations that can be used with minimal training time and a small amount of labeled data for a wide range of Earth monitoring applications.

Building a model capable of providing multitask representations is in line with the recent DL paradigm known as "foundation model" (FM) [5]. In remote sensing (RS), FMs have been recently theorized as a solution to avoid the training of numerous task-specific models, exploiting the synergy of multimodal SITS, as well as mitigating the need for costly and time-consuming data annotation [6]. In fact, over the past years, a wide variety of so-called FMs have been proposed [7], [8], [9], [10], [11], [12]. While all these models are pretrained using self-supervised learning (SSL) strategies, which reduces the need for labeled data, they differ in several important aspects. Among other things, several FMs [8], [9] are evaluated on nondense downstream tasks, which are not relevant to real-world requirements. Besides, certain methods consider either really short time series [7] or, alternatively, do not harness temporal information [10], [11]. As the aim of this article is land monitoring, we consider it necessary to develop an FM capable to take into account the specificities of SITS, which are among others *irregular* and *unaligned*. More precisely, irregularity refers to variable temporal gaps within the same time series, caused by missing acquisitions due, for example, to inappropriate atmospheric conditions (haze, fog,

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or clouds) or sensor problems. Unalignment corresponds to different acquisition dates when comparing at least two time series, due to orbital phase shifts. To remove any ambiguity, we consider that for land monitoring, an FM should process multimodal SITS and generate: 1) *generic*; 2) *easy-to-use*; and 3) *informative* representations suitable for large-scale mapping applications. Given the numerous challenges involved in building an FM, this article does not examine the processing of multimodal data. We detail the three remaining properties as follows.

1) *Providing Easy-to-Use Representations of Irregular and Unaligned SITS:* FMs should provide representations that serve as a basis for the geosciences and climate communities. As a consequence, we consider that easy-to-use representations are relevant without model fine-tuning, are aligned and of fixed size, and preserve the spatial resolution of the SITS. In this article, the term "aligned" is used to emphasize that, even if the input SITS are initially unaligned, the proposed method should generate comparable representations. No existing SITS representation encoder [9], [13], [14], [15] matches the three criteria. For instance, most current methods [8], [9], [13], [14] provide representations of SITS whose temporal dimensions match those of the input SITS. As a result, these methods provide unaligned representations of variable temporal size. Therefore, such representations cannot be directly used in traditional machine learning (ML) methods for regression or classification tasks. Another study [16] proposes to collapse the temporal dimension of the latent representation during training. Nevertheless, the proposed latent representation also has a lower spatial resolution than the input, unsuitable for full-resolution mapping.
2) *Learning Informative Representations From Unlabeled Data:* The SSL pretraining strategy used to train an FM should yield meaningful high-level SITS representations. Masked autoencoders (MAEs) have been frequently employed for SITS pretraining due to their ease of implementation [9], [13], [14], [15]. SITS encoders are trained to reconstruct parts of the input SITS. Nevertheless, the ability of MAE to extract high-level features has been questioned [17]. Indeed, it is hypothesized that being trained to predict in a low-level semantic space (pixel level) could reduce the ability to learn more complex and abstract SITS representations. In addition, other SSL training techniques compute a loss at the level of the latent space, which is supposed to be of a higher semantic level than the input data. For example, instance discrimination strategies are multiview SSL techniques designed to maximize the similarity between representations of two views from the same input data. Views of the input data must be generated in such a way as to preserve the semantic information needed to solve downstream tasks. While artificial data augmentation is often employed to generate views in computer vision, such techniques are not relevant for SITS. Apart from SkySense [8], instance discrimination remains largely unexplored in SITS because it requires domain-specific view generation and often needs large batch sizes. To mitigate the disadvantages of both types of SSL strategies, recent works propose hybrid approaches [18], [19]. These methods combine various SSL strategies, such as instance discrimination with MAE, to learn more informative representations. Except for [16], these hybrid approaches are not applied to SITS representation learning.
3) *Providing and Assessing Generic Representations:* The genericity of an SITS representation refers to the ability of a model to perform well on two important scenarios: first, in a variety of geographic and temporal configurations, and second, in a variety of downstream tasks. Under this purpose, it is crucial to pretrain FM on large-scale datasets covering diverse geographical and temporal configurations, which avoids learning representations only relevant to specific areas and periods. Nevertheless, there is currently a significant lack of well-designed large-scale unlabeled datasets for SITS. Existing datasets [7], [20], [21] do not provide enough temporal acquisitions or correspond to restricted geographical and temporal configurations [22]. In addition, the production of RS FM is also limited by the lack of downstream reference tasks that are representative of the needs of the geosciences and climate communities. While several FMs [9], [21] in RS are evaluated on scene-classification tasks, most real-world RS applications necessitate high spatial resolution semantic maps. Despite some growth, downstream labeled segmentation datasets for SITS remain scarce, limiting the evaluation of FM in the production of generic spectro-spatio-temporal representations of SITS.

In view of 1)–3), we propose an ALIgned Sits Encoder (ALISE) as a further step toward the development of an FM for SITS. As the numerous criteria mentioned above already require substantial research, ALISE was designed to process data from just one sensor (S2). Although ALISE is not an FM, it takes steps toward the production of easy-to-use, informative, and generic representations. First, ALISE furnishes aligned and fixed-size representations, leveraging the spatial, spectral, and temporal dimensions of multiyear SITS, which are easy to use [see 1)]. The resulting ALISE representations also preserve the spatial resolution of the input SITS, which is considered as crucial for downstream segmentation tasks. Second, hybrid SSL strategies are investigated to obtain informative SITS representations [see 2)]. Notably, we have studied the possibility of integrating an SSL instance discrimination strategy alongside an MAE task. We propose a cross-view reconstruction task, where views are subseries of the original SITS, which are each composed of different acquisitions. We also investigate whether integrating additional instance discrimination losses leads to more informative latent representations. These losses enforce invariance between the SITS views' representations and decorrelate latent variables. Third, generic representations are sought by using a new pretraining unlabeled open-source dataset and evaluating the model on three downstream tasks [see 3)]. In particular, the three distinct downstream tasks are: crop segmentation (PASTIS [3]), dense land cover segmentation (MultiSenGE [23]), and change detection (with the specially designed *CropRot* dataset [24]).

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This novel labeled dataset was built to enhance the benchmark of downstream tasks for the assessment of FMs. On the two segmentation downstream tasks, we train a single linear layer to perform pixel-level classification. The quality of ALISE's representations is evaluated by using them in frozen (linear-probing) configurations. Finally, the change detection task is performed without any additional learning step. Change maps are generated by measuring the distance between a pair of aligned SITS representations obtained from ALISE. In addition, this article details: ALISE's performance under a labeled data scarcity scenario, a qualitative assessment of the proposed temporal alignment method, and an extensive study of the influence of the view generation protocol and instance discrimination losses.

Our contributions can be summarized as follows.

1) We introduce ALISE, a novel SITS encoder that provides aligned representations of SITS at high spatial resolution.
2) We explore a new multiview SSL task specifically designed for SITS.
3) We provide two novel datasets: a large-scale unlabeled pretraining multiyear European S2 dataset [25] and a labeled downstream crop change detection dataset [24].
4) We achieve state-of-the-art (SOTA) performance on linear-probing segmentation tasks [9], [15].

The code[1] and the pretraining [25] and change detection [24] datasets are available. The remainder of this article is organized as follows. First, Section II presents the current state of the art in terms of SSL strategies for SITS, methods for aligning irregular and unaligned SITS, and pretraining datasets. Next, Section III corresponds to our proposed methodology. Moreover, the explanation of the experimental setup is detailed in Section IV and the results are detailed in Section V. Finally, conclusions are drawn in Section VI.

II. Related Works

In this section, we review works related to the production of informative (Sections II-A and II-B), ready-to-use (Section II-C), and generic (Section II-D) representations. More specifically, this section presents an overview of the two main categories of SSL, which are MAE applied in most cases with SITS (Section II-A), and instance discrimination SSL strategies, which have rarely been explored in the context of SITS (Section II-B). Subsequently, methodologies for the generation of aligned SITS representations are presented (Section II-C), followed by an overview of existing S2 large-scale pretraining datasets (Section II-D).

A. MAEs for Time Series

MAEs were popularized thanks to the great performance obtained by BERT [26]. The MAE strategy involves corrupting multiple elements (tokens) of the input sequence and training the model to reconstruct them. This SSL strategy was soon extended to other domains such as time series [27], [28], [29] or image processing [30]. In the field of RS, a variety of studies employed MAE in the analysis of monodate satellite images [11], [31] or satellite video [12]. For SITS, the proposed MAE uses either a temporal masking strategy with a temporal transformer or a spatiotemporal masking strategy with a Vision Transformer (ViT). As detailed in [15], existing spatiotemporal masking strategies, such as the SatMAE [21] and Prithvi [7], process exclusively SITS composed of three acquisitions. The temporal approaches, on the other hand, are often applied on fully temporal architectures [9], [13] or with narrow spatial context [14]. To the best of our knowledge, the sole spatio-spectro-temporal architecture adapted to SITS and pretrained as an MAE is U-BARN [15]. Furthermore, MAE methodologies adapted to time series (not specifically SITS) differ on two important points.

First, they differ on how they handle corrupted tokens. Inspired by the original BERT [26], methods with temporal masking inject the corrupted tokens directly into the SITS encoder. Unfortunately, this input data corruption introduces a distribution shift between pretraining and downstream tasks, as the latter have no input corruption step. Spatiotemporal methods [7], [21] address this last limitation by using an asymmetric encoder–decoder architecture. Inspired by MAE in vision [30], the corrupted tokens are not fed to the encoder. Instead, the latter are concatenated to the input representations and processed solely by the decoder using a self-attention mechanism. In addition, outside RS studies, recent MAEs for time series such as [28] and [29] also employ an asymmetric architecture. Instead of a self-attention mechanism, a lightweight decoder that performs cross-attention between corrupted tokens and the latent representation is employed.

Second, MAE on time series differs in the employed masking pattern. While retrieving a masked word in natural language processing (NLP) requires a holistic understanding of the sentence, neighboring data points in time series are often highly correlated. Therefore, several studies [27], [28], [29] advocate splitting the time series into nonoverlapping temporal subseries before masking. Therefore, the masking strategy is applied at the subseries level to force the model to reconstruct local variations. However, this methodology is not directly applicable to irregular SITS, where each subseries would represent different temporal scales. Existing methods on SITS thus often mask random time steps [14], [15], ignoring the potential redundancy of the temporal information. Sole [9] considers consecutive acquisition masking.

Consequently, unlike several previous studies on SITS [13], [14], [15], [21], we propose a temporal masking strategy, where the corrupted tokens are not processed by the SITS encoder. We also consider masking successive acquisitions, and the reconstruction task utilizes a lightweight decoder with cross-attention.

B. Self-Supervised Instance Discrimination Methods

The use of MAE in vision has nevertheless been criticized for focusing on learning local relationships within an input sample, instead of modeling the relationship between samples [32]. Besides, while MAE techniques are easy to implement, they can produce representations that are generally of a lower semantic level than instance discrimination techniques [33].

[1] https://src.koda.cnrs.fr/iris.dumeur/alise

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As per [34], we consider instance discrimination as a subset of SSL, where a Siamese network (composed of two branches) is trained to produce similar representations of two views of the same data. The views correspond to alternative ways of observing the input data. Different views can be, for example, signals from different sensors or artificially augmented versions of the input data. In addition, views are expected to preserve the input semantic information needed for downstream tasks. While computer vision studies often employ a set of benchmark data augmentation techniques, determining relevant view generation on SITS is challenging. Nevertheless, several studies have tackled view generation methods on single RS image analysis. For instance, Mañas et al. [35] have shown the interest of using domain-adapted augmentation on RS images. These methods propose to consider two images taken at the same location but at different times as two views of the same input data. Other works exploit the multimodality and consider images from different sensors (optical and radar pairs) as two different views [36], [37].

Multiview SSL techniques can be divided into four categories: contrastive [38], clustering [39], [40], distillation [41], [42], [43], and redundancy reduction [44], [45], [46]. These approaches differ in their strategies to prevent representation collapse, a scenario in which the encoder always predicts the same representation regardless of the input.

Contrastive learning [38] and its variants for segmentation tasks [47] heavily rely on negative pair sampling (i.e., finding pairs of samples representing different semantics). Efficient negative pair sampling is challenging for SITS because pixels from different SITS may still represent the same classes. In the absence of labels, it is difficult to identify negative examples that are not trivial and are actually beneficial to learning. Consequently, for pixel-level SITS classification, a contrastive loss is often used in a semi-supervised framework where labels help generate relevant negative pairs [48]. SSL clustering methods, on the other hand, do not require sampling of negative pairs. Instead, they use a clustering algorithm dedicated to the generation of pseudo-labels (also called prototypes). To do so, strong assumptions about the batch distribution are required. For example, Caron et al. [40] assume that all examples in a batch are evenly distributed among the prototypes. Distillation-based techniques [41], [42], [43], on the other hand, require complex training tricks such as momentum encoder or stop gradient. Finally, compared to other instance discrimination SSL frameworks, the implementation of redundancy reduction techniques [44], [45], [46] is straightforward. These strategies prevent informational collapse by decorrelating every pair of variables of the embedded latent representation. The VicReg approach [45] proposes the use of three losses: the invariance loss, which enforces similarity between the embedded latent representations of the two views; the variance loss, which maintains the variance of the embedded variables above a threshold; and the covariance loss, which intends to decorrelate the variables of each embedded view. Furthermore, a modified version of VicReg, named VicRegL [46], has been adjusted for downstream segmentation tasks. In this latter method, the three previous losses are also calculated at the pixel level. Consequently, due to its simplicity, the combination of VicReg losses with a cross-reconstruction multiview task is explored in this article.

C. Processing Irregular and Unaligned SITS

For downstream tasks, SITS representations provided by pretrained models must be of fixed size, in order to be injected into classical lightweight classifiers, including Random Forest [49], TempCNN [4], recurrent neural networks [50], and sparse variational Gaussian process (SVGP) [51]. Additionally, those representations should be "aligned," enabling the direct comparison of the features of two different samples.

The generation of aligned SITS prior to their use in an ML model has been addressed in several works. A common practice is to perform a linear interpolation of the annual time series into a common temporal grid [49]. While this method is efficient for annual SITS classification, it might remove fine-grained information or introduce noise. Besides, cloud masks are required to perform the interpolation, and the definition of reference dates, composing the common temporal grid, is challenging. Another common preprocessing method is the generation of *composite* images, which aims to summarize valid acquisitions over a temporal extent. For example, Tseng et al. [9] suggest using time series with monthly information to fuse multisensor SITS. In this case, optical SITS monthly information corresponds to the least cloudy image, while a median of synthetic aperture radar (SAR) acquisitions over a month is used. Unfortunately, this temporal preprocessing strategy to prevent irregular temporal feature representations has several shortcomings. First, the proposed monthly optical sampling protocol does not ensure that each pixel of the image has a clear acquisition. Second, the monthly sampling protocol induces an important loss of temporal information.

Furthermore, the latter two methods are not flexible and do not adjust to the target task. For this reason "data-driven alignment strategies" have been proposed. For instance, Bellet et al. [52] propose the learning of an attention-based interpolation (mTAN) [53], which is trained end-to-end along a Gaussian process classifier (SVGP). In particular, the interpolation weights are computed thanks to a scaled dot product between embedded acquisition dates and embedded reference dates. The temporal embedding of the dates is performed with learnable embedding functions [54]. Besides, Bellet et al. [52] show that the attention-based interpolation framework outperforms SVGP fed with linearly interpolated data. Although this mechanism is more flexible and relevant for a classifier requiring aligned SITS such as SVGP, novel attention-based encoder architectures can now process irregular and unaligned SITS [55]. With these latter networks, a temporal preprocessing will result in an unnecessary loss of information. As a consequence, aligning SITS after their encoding by an attention-based encoder seems a more appropriate approach. A second limitation of mTAN is that the interpolation weights are calculated exclusively with the acquisition dates, without consideration of the content of the time series.

Recently, in computer vision, advanced DL architectures such as the Perceiver I/O [56] have emerged to map arbitrary input sequences onto a fixed-size aligned latent space using a

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TABLE I

DESCRIPTION OF S2 SITS UNLABELED TRAINING DATASETS USED TO PRETRAIN LARGE-SCALE SITS MODELS. TEMPORAL EXTENT REFERS TO THE LONGEST TIME INTERVAL EXISTING IN THE DATASET. THE NUMBER OF DATES CORRESPONDS TO THE EXACT OR AVERAGE (∼) NUMBER OF DATES IN THE SITS. THE ROI SIZE CORRESPONDS TO THE SPATIAL DIMENSIONS OF THE IMAGES WITHIN THE SITS. AVAILABLE INDICATES WHETHER THE DATASET IS AVAILABLE FOR DOWNLOAD. CLOUD FILTER INDICATES WHETHER CLOUDY IMAGES HAVE BEEN REMOVED. WHEN THE PERCENTAGE IS SHOWN, IT SPECIFIES THE MAXIMUM PERCENTAGE OF CLOUDS ACCEPTED IN AN IMAGE. N/A IS EMPLOYED WHEN THE INFORMATION IS NOT GIVEN IN THE CORRESPONDING ARTICLE

Data-Set Name	Data	Temporal Extent	Number of dates	Geographical extent	Roi size	Available	Cloud filter
SSL4EO-S12 [20]	S2-L2A 13 bands	2020	4, (1/season)	Worldwide	264×264	✓	yes, $\leq$ 10 %
FR-S2 (U-BARN) [15]	S2-L2A 10 bands + valid mask	2018-2020	~ 179	France	1024×1024	✓	yes, $\leq$ 30 %
Presto [9]	S2-L2A 10 bands	2020-2021	24 (1/month)	Worldwide	1×1	✗	yes
Prithvi [7]	NASA HLS[2]V2 L30	N/A	N/A	USA	64×64	✗	yes
SkySense [8]	S2 L2A 10 bands	N/A	~ 65	Wordlwide	$64 \times \times 64$	✗	yes, $\leq$ 1 %
Clay[3]	S2 L2A 10 bands	2018-2023	8, (1/quarter)	Worldwide	224×224	✓	yes
MMDC EU (ours)	S2 L2A 10 bands + valid mask	2017-2020	~ 354	Europe	512×512	✓	no

flexible cross-attention querying mechanism. Specifically, the projection of the input sequence is determined by a scaled dot product between the input sequence and a learnable array, denoted *learnable queries*. The resulting length of the output aligned sequence is determined by the number of learned queries. This mechanism has been applied to SITS with a single learnable query in [16] to provide SITS representations with a collapsed temporal dimension. However, the use of this mechanism with a larger number of queries and its analysis has not yet been investigated.

As a result, we propose a flexible query mechanism to align irregular and unaligned SITS after their encoding by an attention-based encoder [57]. We also investigate how the information is stored in the aligned latent representations.

D. S2 Pretraining Datasets

As the number of attempts to construct an RS FM increases, a multitude of pretraining datasets have been constructed. Table I provides an overview of existing large-scale pretraining datasets employed to pretrain SITS encoders. Ideally, an RS FM should be trained on multispectral data covering multiple geographic and temporal configurations. The pretraining data should also include SITS with numerous acquisitions to allow learning complex temporal features. Moreover, an SITS FM is expected to learn to ignore cloudy pixels. As a consequence, invalid pixels should not be removed from the pretraining input data. Finally, to train a spatio-spectro-temporal network, it is necessary to collect SITS with a large spatial extent.

With the exception of the Prithvi dataset, all the proposed datasets provide at least the ten S2-bands (10–20-m resolution). In addition, except for FR-S2, the datasets show remarkable geographical variability. However, none of the existing datasets fits all the aforementioned criteria to pretrain a large-scale SITS FM. Indeed, most of these datasets [7], [9], [20] do not provide long enough time series to capture the complex temporal dynamics of the Earth's surface. Additionally, several pretraining datasets employ a severe cloud filtering of the information (SSL4EO-S12, Presto, and Clay datasets). As a consequence, this filtering prevents the FM from identifying cloudy pixels as irrelevant. This could decrease performance on downstream tasks where the cloud mask information may not be available. In contrast, a pretraining dataset like FR-S2 applies less strict cloud filtering and provides validity mask. Besides, it is presumed that an RS FM pretraining dataset should be balanced [6]. Nevertheless, some existing datasets, such as SSL4EO-S12 [20], focus on urban areas. The scarcity of natural landscapes in such datasets could be an obstacle to learning complex temporal dynamics.

Consequently, we introduce a novel large-scale European S2 SITS pretraining dataset named MMDC-EU, which is detailed in Section IV-A1.

III. METHOD

The method consists in pretraining an ALISE, which produces aligned representations for multiyear irregular and unaligned SITS. The details of the ALISE architecture are presented in Section III-A, followed by the description of the multiview self-supervised pretraining strategy. Specifically, the proposed SSL method studies the combination of two types of losses: a generative cross-reconstruction loss and instance discrimination losses computed in the latent space.

A. ALISE: ALIgned Sits Encoder

ALISE harnesses the spectral, spatial, and temporal dimensions of irregular and unaligned input time series

[2]Harmonized Landsat Sentinel 2 dataset.

[3]https://clay-foundation.github.io/model/release-notes/data_sampling.html

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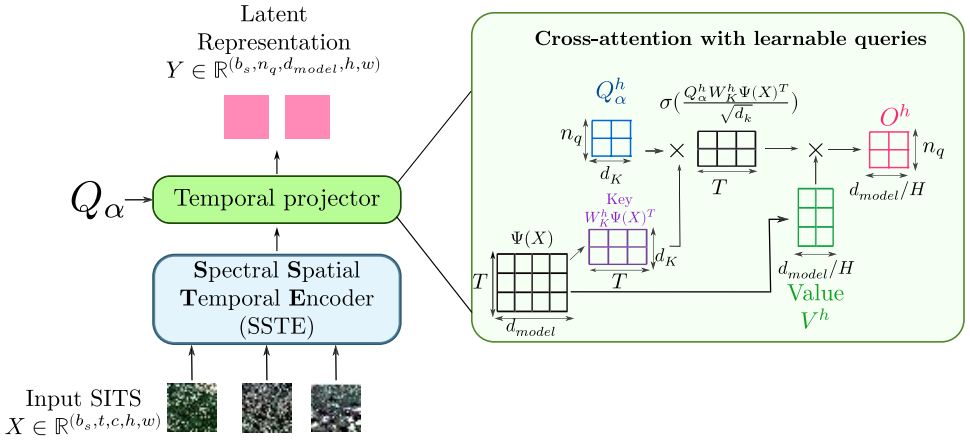


Fig. 1. Overall description of ALISE architecture. The input time series X is first processed by the SSTE [15]. The obtained intermediate representations, noted $\Psi(X)$, are then processed by a temporal projector. The temporal projector corresponds to a multihead cross-attention mechanism with learnable queries Q_α. For visual clarity, the cross-attention is represented for one attention head.

$X \in \mathbb{R}^{(b_s,t,c,h,w)}$, where b_s, t, and c are, respectively, the batch, temporal, and spectral dimensions and h and w the spatial dimensions. Although t may vary for each SITS, ALISE generates a latent representation $Y \in \mathbb{R}^{(b_s,n_q,d_{\text{model}},h,w)}$ of fixed dimensions, where d_{model} and n_q are the channel and temporal sizes, respectively.

As illustrated in Fig. 1, ALISE is composed of two main blocks. First, a spatial–spectral–temporal encoder (SSTE), noted Ψ in (1), which corresponds to the U-BARN architecture detailed in [15] and Appendix C. As the original U-BARN was initially designed to handle annual SITS, the positional encoding (PE) in ALISE has been modified to process multi-year SITS. Specifically, the temporal information provided to ALISE is not the day of year (DoY). Instead, δ_t, which is the difference in days between the image acquisition date and a given reference date (March 3, 2014), is employed

$$O^h = \sigma\left(\frac{Q^h_\alpha W_1^{h^T} \Psi(X)^T}{\sqrt{d_K}}\right) V^h. \tag{1}$$

Besides, a temporal projector processes the irregular and unaligned outputs of the SSTE, $\Psi(X)$, to generate aligned SITS representations. The projection is performed by a multi-head temporal cross-attention mechanism between learnable queries and $\Psi(X)$. The scaled dot product of the cross-attention mechanism on a head h is detailed in (1) with Q^h_α the learnable queries, X the input time series, d_{model} the number of features in $\Psi(X)$, H the number of heads, and σ the softmax function. The attention product is fully temporal, and thus, $Q^h_\alpha \in \mathbb{R}^{(n_q,d_K)}$, $W^h_1 \in \mathbb{R}^{(d_{\text{model}},d_K)}$, and $\Psi(X) \in \mathbb{R}^{(t,d_{\text{model}})}$. The values $V^h \in \mathbb{R}^{(T,d_{\text{model}}/H)}$ correspond to a split across the channel dimensions of the output of the SITS $\Psi(X)$. As detailed in (2), all the outputs of each head $O^h \in \mathbb{R}^{n_q,d_{\text{model}}/H}$ are then concatenated along the feature dimension and processed by a multilayer perceptron (MLP), which generates the latent representation Y

$$Y = \text{MLP}\big(\text{concat}_h\big(O^1, \ldots, O^H\big)\big). \tag{2}$$

The temporal dimension of the latent representation Y is determined by the number of learnable queries (n_q). It must be noted that the temporal projector does not shrink the spatial dimension of $\Psi(X)$, meaning that each pixel of the SITS is represented by d_{model} features along n_q temporal features. It is crucial to understand that in the resulting latent representations, the notion of "time" is not preserved. In other words, the time series is folded in a way that does not preserve the notion of order or distance in the time axis. As a result, the latent representation is not a time series; rather, it is a stack of temporal features. Therefore, we refer to the n_q vectors in the aligned latent representation as the "latent temporal features."

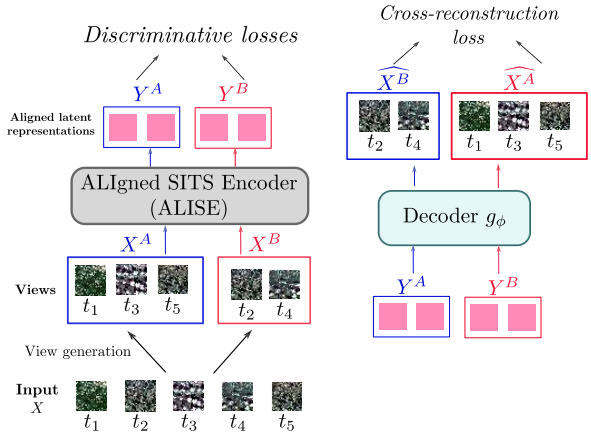


Fig. 2. Description of the proposed multiview SSL strategy. Given an input time series X, two views are generated: X^A and X^B. Each view is processed independently by ALISE, which generates their respective aligned latent representations Y^A and Y^B. A decoder g_ϕ is trained to reconstruct one view using the latent representation of the other. Additional discriminative losses are computed on the latent representation.

B. Multiview Pretraining Task

As detailed in Fig. 2, the multiview SSL pretraining task combines a cross-reconstruction loss with two additional losses computed on the embedded latent representations. As detailed in (3), the total SSL loss corresponds to the weighted sum of three terms L_{inv}, L_{cov}, and L_{rec}, respectively, the

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invariance, covariance, and reconstruction losses

$$L = w_{\text{inv}} L_{\text{inv}} + w_{\text{cov}} L_{\text{cov}} + w_{\text{rec}} L_{\text{rec}}. \tag{3}$$

In summary, L_{rec} allows ALISE to be pretrained in the same manner as MAEs, which have shown encouraging results on SITS [9], [14], [15]. In addition, L_{inv} and L_{cov} correspond to instance discrimination losses, which are used in an attempt to resolve the difficulty of MAEs in generating high-level semantic representations.

1) View Generation: After the view generation phase, ALISE encodes each of the two views, resulting in two aligned representations. The latter are used to compute the invariance and covariance losses. In the cross-reconstruction loss, the representation of each view is used to reconstruct the other view. The objective of the view generation protocol is to provide views that preserve semantic content. For SITS, the generation process aims to create views that maintain the pixel information of the observed landscape. Consequently, the two views, X^A and X^B, represent a time series at the same location but with different acquisition times. The view generation process starts by selecting N adjacent acquisitions among an irregular and multiyear SITS. As detailed in (4) and Fig. 3, this latter time series is divided into n_w nonoverlapping temporal windows, each composed of t_w dates. Given that SITS is irregular, each temporal window may represent a different temporal span

$$X = \bigcup_{i=0}^{n_w - 1} \left\{ X_j \mid i \times t_w \leq j < (i+1) \times t_w \right\}. \tag{4}$$

Finally, to ensure that the two views cover nearly identical periods, every other temporal window is used to construct respectively X^A [see (5a)] and X^B [see (5b)]. Therefore, t_w corresponds to the number of consecutive dates that the model has to predict during the training process. We posit that increasing t_w complexifies the cross-reconstruction task as more temporal variations have to be retrieved by the model. This generation approach ensures that the views are temporally intertwined: $X^A \cup X^B = X$ and $X^A \cap X^B = \emptyset$ and provides a parameter t_w, which controls the difficulty of the pretraining task

$$X^A = \bigcup_{i=0}^{\frac{n_w}{2} - 1} \left\{ X_j \mid 2 \times i \times t_w \leq j < (2 \times i + 1) \times t_w \right\} \tag{5a}$$

$$X^B = \bigcup_{i=0}^{\frac{n_w}{2} - 1} \left\{ X_j \mid (2 \times i + 1) \times t_w \leq j < (2 \times i + 2) \times t_w \right\}. \tag{5b}$$

2) Discriminative Losses: As illustrated in Fig. 2, the augmented views X^A and X^B are independently encoded by ALISE. The aligned latent representations Y^A and Y^B are then processed into embeddings Z^A and Z^B by a projector. The projector aims to eliminate the information by which the two representations differ. Specifically, the projector operates exclusively on the channel dimensions: $\pi_\omega : \mathbb{R}^{(d_{\text{model}})} \to \mathbb{R}^{(d_{\text{emb}})}$. In other words, pixel-level latent vectors of each n_q query are independently processed by the projector. As shown in Fig. 4,

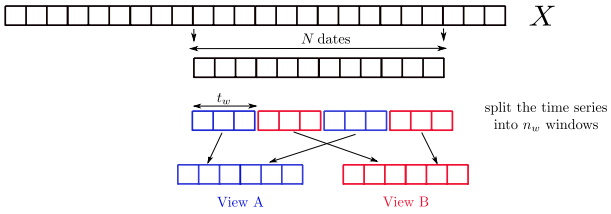


Fig. 3. Illustration of the view generation protocol with $N = 12$, and $t_w = 3$.

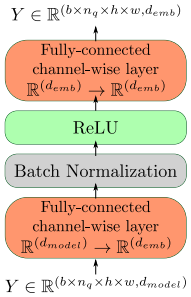


Fig. 4. Description of the projector architecture.

the proposed projector consists of one fully connected layer followed by batch normalization and ReLU, and a second linear layer. It is assumed that the choice of the projector's architecture affects the computation of the covariance loss. However, no empirical benefits were found from using a deeper or wider projector architecture for our considered downstream tasks.

We denote $\mathbf{z}^k_{(b,n,i,j)} \in \mathbb{R}^{d_{\text{emb}}}$ the pixel-level embedded vector of Z^k located at the spatial position (i, j) for the nth query and bth batch position. We propose to compute the invariance and covariance losses on the embeddings Z^A and Z^B.

First, the invariance loss maximizes the similarity between the embedded vectors $\mathbf{z}^A$ and $\mathbf{z}^B$ [see (6)]. As X^A and X^B have distinct acquisition dates but cover the same time period, L_{inv} aims at learning representations that are invariant to the acquisition dates

$$L_{\text{inv}}(Z^A, Z^B) = \frac{1}{b_s \times n_q \times h \times w} \sum_{(b,n,i,j)} \left\| \mathbf{z}^A_{b,n,i,j} - \mathbf{z}^B_{b,n,i,j} \right\|_2^2. \tag{6}$$

Second, we also investigate whether the covariance loss allows learning better representations. The covariance loss decorrelates the different d_{emb} features. The total covariance loss [see (8)] corresponds to the sum of the covariance losses computed for each embedding Z^k. For centered embeddings $Z \in \mathbb{R}^{(b_s \times n_q \times h \times w, d_{\text{emb}})}$, the covariance loss aims to minimize the off-diagonal values of the covariance matrix $C(Z)$ in (7). In other words, the covariance matrix of the d_{emb} variables is estimated on a batch composed of $b_s \times n_q \times h \times w$ samples. In Appendix B, we discuss how these discriminative losses are related to the VicRegL [46] losses

$$l_{\text{cov}}(Z) = \frac{1}{d_{\text{emb}}} \sum_{i \neq j} [C(Z)]_{i,j}^2 \tag{7}$$

$$L_{\text{cov}} = l_{\text{cov}}(Z^A) + l_{\text{cov}}(Z^B). \tag{8}$$

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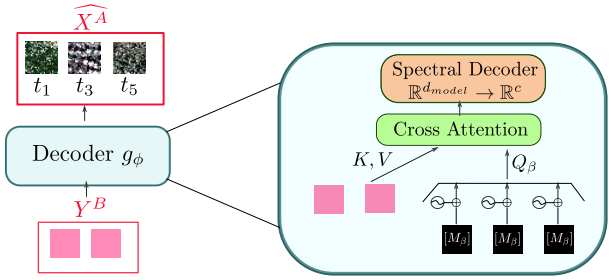


Fig. 5. Description of the lightweight decoder employed for the cross-reconstruction task. The spectral decoder is a fully connected layer.

3) Cross Reconstruction Task: As depicted in Fig. 5, the latent representations Y^A and Y^B are employed in a cross-reconstruction task. A specific fully temporal decoder using a cross-attention mechanism followed by a fully connected layer is trained to recover one view X^B (resp. X^A) from the latent representation Y^A (resp. Y^B) of the other view X^A (resp. X^B). The fully connected layer operates exclusively on the channel dimension of each pixel of the images, to recover the S2-bands from the d_{model} features.

As proposed in [28], the cross-attention mechanism exploits $Q_\beta \in \mathbb{R}^{(t_w \times n_w, d_{\text{model}})}$, which specifies the dates to be reconstructed. As detailed in (9), Q_β corresponds to the sum of a shared learnable masked token $M_\beta \in \mathbb{R}^{(d_{\text{model}})}$ with the temporal PE[4] of the date to reconstruct. Additionally, as described in (10), the latent representation Y^k with $k \in \{A, B\}$ is used to construct the keys $Y^k W_2$ and the values Y^k

$$Q_\beta = \left[M_\beta + PE\left(\delta_{t_i}\right)\right]_{1 \le i \le t_w \times n_w} \tag{9}$$

$$\text{Cross Attention}\left(Q_\beta, Y^k\right) = \sigma\left(\frac{Q_\beta W_1 W_2^T Y^{k^T}}{\sqrt{d_{\text{model}}}}\right) Y^k. \tag{10}$$

Finally, the quality of the reconstruction is assessed by using the classical mean square error. As described in (11), the reconstruction loss is the average of the reconstruction losses of each view

$$L_{\text{rec}} = \frac{1}{2}\left[l_{\text{rec}}\left(X^A, Y^B\right) + l_{\text{rec}}\left(X^B, Y^A\right)\right]. \tag{11}$$

Following the approach of [15], pixels with invalid measurements due to the acquisition conditions (e.g., cloudy and out-of-swath pixels) are ignored in the reconstruction loss. As detailed in (12), M_t^{valid} represents the Boolean validity mask, n_t^{valid} represents the number of clear pixels, $T = (n_w \times t_w/2)$ represents the number of acquisitions in a view, and $\odot$ is the Hadamard product

$$l_{\text{rec}}\left(X^k, Y^l\right) = \frac{1}{T}\sum_{t=1}^{T} \frac{M_t^{\text{valid}}}{n_t^{\text{valid}}} \odot \left\|X_t^k - g_\phi\left(Y^l\right)_t\right\|_2^2. \tag{12}$$

The validity mask is only used in the cross-reconstruction loss and is not included in the input data injected into ALISE. Therefore, no validity masks are required for downstream tasks.

[4]The temporal PE used is the same as the one employed in ALISE.

TABLE II
DEFAULT HYPERPARAMETERS FOR PRETRAINING ALISE

t_w	n_q	batch size	d_{model}	d_{emb}	w_{rec}	w_{inv}	w_{cov}	H
2	10	2	64	128	1	1	0	2

C. Implementation Details

To pretrain ALISE, the cosine annealing scheduler with warm restarts [58] was employed with $T_0 = 2$ and a maximum learning rate of $1e^{-3}$. To generate the different views from a multiyear SITS, 60 consecutive dates were randomly selected among the four years of data. Within our unlabeled dataset, 60 consecutive acquisitions can extend over a maximum of four years of data and a minimum of four months. To increase the diversity of the training data, the selection of the N consecutive dates used in the view generation is random for each SITS and changes at each epoch. Besides, ALISE architectural hyperparameters are also detailed in Appendix C. The pretraining tasks were conducted on a single Tesla V100 GPU for 260 epochs. The values of the pretraining hyperparameters are shown in Table II and their choices are explained in Section V-D. The pretrained model with the lowest loss on the pretraining validation set is selected for downstream task assessment.

IV. EXPERIMENTAL SETUP

First, the four S2 L2A datasets used in our different experiments are presented: the novel unlabeled large-scale dataset (MMDC-EU) used for pretraining ALISE and the three downstream labeled datasets (PASTIS, MultiSenGE, and the novel *CropRot*). Second, the implementation details of our two types of downstream tasks setup (semantic segmentation and change detection) as well as the corresponding competitive works are described.

A. Datasets

ALISE is pretrained on a large-scale multiyear European dataset. Besides, three labeled datasets are used to assess the quality of the pretrained SITS encoders. The geographical distribution of the different datasets used is presented in Fig. 6. For these datasets, only the four 10 m and the six 20-m resolution bands of S2 are used. The 20-m resolution bands are resampled onto the 10-m resolution grid by bi-cubic interpolation. Similar to [15], a robust data normalization is applied on S2 L2A reflectances. Due to GPU memory limitations, ALISE is trained to process SITS with a spatial dimension of 64 × 64 pixels (at 10-m resolution). If the used dataset provides larger images, a random crop[5] (resp. center crop) is operated during training (resp. validation/testing) steps.

1) MMDC-EU: We have constructed an unlabeled, multiyear, multimodal SITS dataset spanning Europe. This multimodal datacube is designated as MMDC-EU. In practice, this dataset is composed of the following data: the S2 L2A product, the Sentinel-1 (S1) ascending and descending

[5]https://pytorch.org/vision/main/generated/torchvision.transforms.RandomCrop.html

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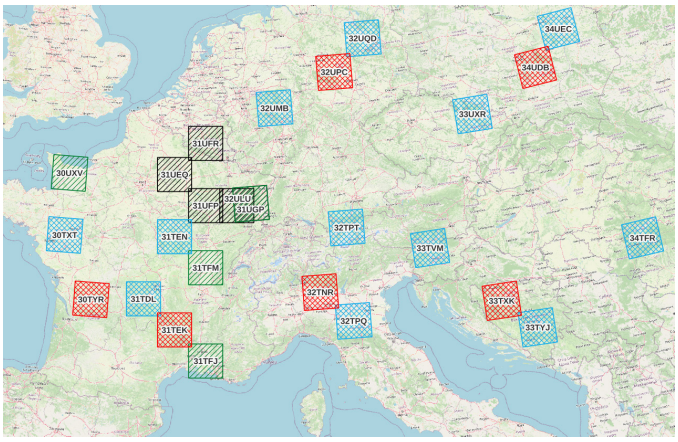


Fig. 6. Geographical distributions of the different tiles composing the datasets. The unlabeled pretraining dataset is composed of multiyear SITS selected within the blue and red boxes for the training and validation datasets, respectively. MultiSenGE labeled data are selected in the area delineated by the black boxes. The PASTIS as well as CropRot datasets are within the green boxes.

acquisitions, ECMWF AGERA5[6] weather variables, and the Copernicus 30 digital elevation model (DEM). Each SITS of each modality is spatially resampled onto the S2 grid. The data cube was downloaded with the openEO platform.[7] The code[8] used to create the multimodal dataset is provided for reference, allowing for potential future expansion. As this article proposes a model that processes exclusively S2 SITS, we focus on the description of the pretraining data for this sole modality. Multiyear S2 SITS from January 2017 to December 2020 was built using all the available acquisitions. The data are split into training and validation sets with, respectively, 1920 and 180 SITS of spatial dimension 64 × 64 pixels. The downloaded S2 SITS corresponds to data processed by Sen2Cor [59]. The validity mask employed in the cross-reconstruction task is built thanks to the information provided by SCL and CLM layers.[9] Specifically, as shown in Fig. 6, the pretraining dataset gathers data from 18 S2 tiles. To build the training dataset, ten smaller regions of interest (ROIs) of size 512 × 512 pixels are randomly selected from each of the 12 training tiles. The disjoint validation dataset is composed of the remaining 6 S2 tiles, from which 30 ROIs of size 128 × 128 pixels are randomly drawn.

2) PASTIS Crop Segmentation: The PASTIS dataset [3] provides labels for 18 crop classes from French Land Parcel information System. The SITS considered in our experiments is collected from January to December 2019. The complete dataset contains 2433 SITS and it is divided into five stratified folds. In line with [15], the segmentation task is performed exclusively on known crop classes. Background and void classes are ignored. The competitive method Presto requires cloud masks. As these data are not available in the original PASTIS dataset, the raw S2 L2A and their cloud masks were downloaded from the Sentinel Hub collection.[10] These S2 data also preprocessed by Sen2Cor are used to assess not exclusively Presto but all models.

3) MultiSenGE Land Cover Segmentation: MultiSenGE [23] is a dense land cover labeled dataset for eastern France in 2020. It is composed of five urban classes and nine natural classes. This dataset is composed solely of images with less than 10% cloud cover, and no cloud mask is provided. The resulting SITS is composed of 3–14 acquisitions. In contrast to PASTIS, MultiSenGE provides dense labels. In this dataset, we selected 4145 SITS with a spatial dimension of 256 × 256 pixels. A random split is performed to divide the dataset into training (60%), validation (16%), and test (24%). The class distribution is detailed in Appendix D. Finally, in opposition to the two previous datasets, MultiSenGE data are preprocessed with MAJA [60] instead of Sen2Cor.

4) CropRot Crop Change Detection: This article presents a new dataset for detecting abrupt changes between two SITS. Unlike the identification of changes in a time series (break detection), which may be solved by signal-based methods, the proposed task requires a more advanced semantic understanding. Compared with existing bi-temporal change tasks, crop rotation requires the identification and comparison of different events such as crop growth and mowing periods. Therefore, two SITSs from two agricultural seasons need to be compared. More specifically, thanks to the labels provided by *RPGExplorer Crop successions* [61], *CropRot* identifies crop rotations between two consecutive years in France. In particular, the RPGExplorer database provides crop sequence labels based on the Registre Parcellaire Graphique (RPG).[11] Within a sequence (e.g., 2015–2020), parcels are unified (each parcel has a unique identifier).

For this dataset, the following classes were selected based on the RPG labels: rapeseed, cereals, proteaginous, soybean, sunflower, maize, rice, tubers, and grassland. These classes categorize vegetation based on its physiological characteristics and can be identified using RS data. Pixels that are not part of these crops for the two years 2019 and 2020 are considered as background. Then, the label *change* is assigned to pixels that have a different label between 2019 and 2020. Each dataset sample includes S2 L2A SITS for 2019 and 2020, with their corresponding labels. The label tensor has three channels containing crop labels for 2019, 2020, and the change label. In our proposed downstream task, change detection is performed while ignoring background pixels. The SITSs were built using the SITS spatial extent from PASTIS where sufficient labels from the RPGExplorer were available. Due to this specific selection, the crop classes proteaginous, soybean, and tuber do not appear in our dataset. Nevertheless, the code used to build this labeled dataset is published,[12] enabling it to be extended to other regions of France and to other years. These missing classes might be integrated into an augmented version of the dataset. The change matrix between 2019 and 2020 is presented in Appendix A.

[6]https://cds.climate.copernicus.eu/cdsapp#!/dataset/sis-agrometeorological-indicators?tab=overview

[7]https://openeo.cloud/

[8]https://src.koda.cnrs.fr/iris.dumeur/openeo_datasets

[9]https://docs.sentinel-hub.com/api/latest/data/sentinel-2-l2a/

[10]https://www.sentinel-hub.com/

[11]https://artificialisation.developpement-durable.gouv.fr/bases-donnees/registre-parcellaire-graphique

[12]https://src.koda.cnrs.fr/iris.dumeur/modcix

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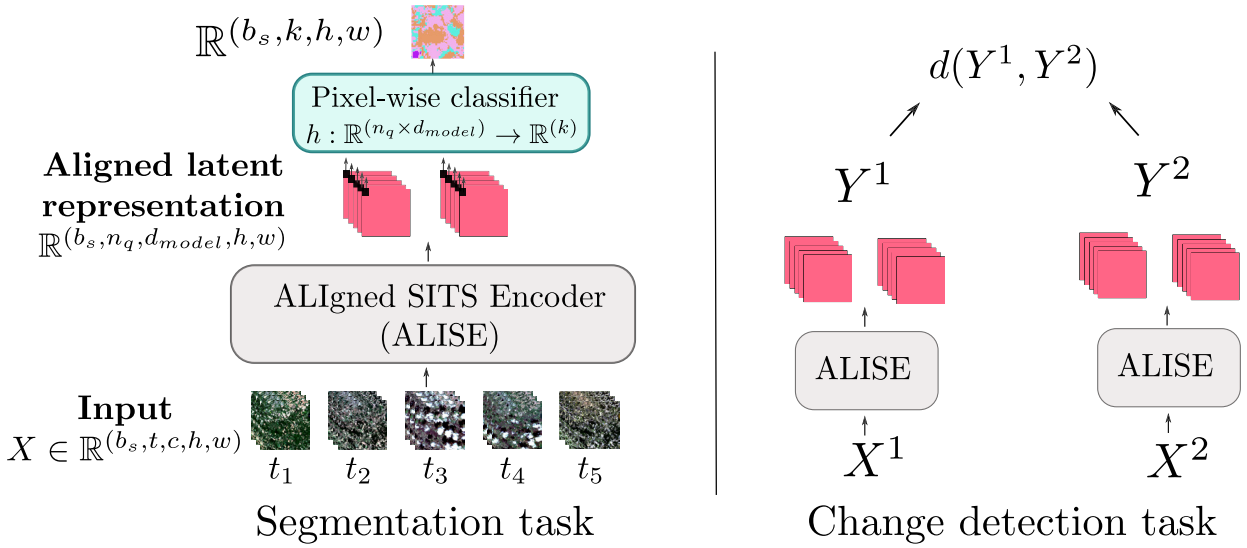


Fig. 7. Two types of downstream tasks considered. (Left) Segmentation task framework. A single fully connected layer projects, for each pixel of the latent representation Y, the $n_q \times d_{model}$ features into a vector of size $\mathbb{R}^k$ with k the number of classes. (Right) Change detection task between two SITS X^1 and X^2. The mean square error is computed between the two aligned latent representations Y^1 and Y^2.

B. Evaluation Protocol

We propose two ways of exploiting ALISE representations, detailed in Fig. 7, corresponding to the two types of downstream tasks: semantic segmentation and change detection.

1) Semantic Segmentation Tasks: As detailed in Fig. 7, we classify each pixel-level latent vector by using a single linear layer in both segmentation tasks. Noting the pixel-level latent vector as $\mathbf{y}_{(b,h,w)} \in \mathbb{R}^{(d_{\text{model}} \times n_q)}$, the unnormalized logits for each class k at the pixel level can be written as: $\mathbf{c}_{(b,h,w)} = \mathbf{y}_{(b,h,w)} A + \mathbf{b}$, where $A \in \mathbb{R}^{(d_{\text{model}} \times n_q, k)}$ and $\mathbf{b} \in \mathbb{R}^k$. The classical cross-entropy loss function is used for training.[13] The latent representations are generated by a pretrained ALISE whose weights are frozen in linear probing or updated during fine-tuning. We denote the fine-tuning and linear-probing configurations as ALISE^{FT} and ALISE^{FR}, respectively, while the fully supervised (FS) model is denoted as ALISE^{FS}. During the downstream tasks, ALISE as well as competitive models are trained with ADAM optimizer, a learning rate of $1e^{-4}$, and ReduceLROnPlateau scheduler with a patience of ten epochs and a decay rate of 0.05.

2) Change Detection: As detailed in (13) and illustrated in Fig. 7, change detection between two SITS X^1 and X^2 is performed at a pixel level. For each pixel located at location (h, w), the mean square error between the corresponding latent vectors $Y^1_{(\cdot,\cdot,h,w)}$ and $Y^2_{(\cdot,\cdot,h,w)}$ is calculated. The resulting distance value serves as change detection criterion

$$d\left(Y^1_{(\cdot,\cdot,h,w)}, Y^2_{(\cdot,\cdot,h,w)}\right) = \frac{1}{n_q \times d_{\text{model}}} \sum_{n,d} \left(y^1_{n,d,h,w} - y^2_{n,d,h,w}\right)^2. \tag{13}$$

C. Competitive Methods

As mentioned above, ALISE is evaluated on two different types of downstream tasks: semantic segmentation and change detection. Consequently, in this section, we detail the competing works associated with these two types of tasks.

TABLE III
DESCRIPTION OF CONSIDERED SSL STUDIES IN RS

Model Name	(i)	(ii)	(iii)	(iv)
DOFA [10]	✗	✓	✓	✓
SpectralGPT [11]	✗	✓	✓	✓
Prithvi [7]	✗	✗	✓	✓
SatMAE [21]	✗	✗	✓	✓
SkySense [8]	✓	✓	✗	✓
OmniSat [16]	✓	✗	✓	✗
Presto [9]	✓	✓	✓	✓

1) SITS Segmentation Concurrent Works: We sought to select competitive models among studies that employed SSL pretraining and met the following criteria.

1) The model processes long SITS (more than three acquisitions).
2) The model is pretrained on a large S2 SITS dataset.
3) The pretrained model is publicly available.
4) The representations are adapted to downstream segmentation tasks.

As indicated in Table III, the only suitable competing work is Presto, as it is the only one to meet all four criteria.

Moreover, as mentioned in Section I, the goal is to produce easy-to-use SITS representations by exploiting the pretrained ALISE model as frozen (FR) in the downstream tasks. Moreover, in our downstream experiments, performances in an FS and fine-tuned (FT) configurations are also provided for comparison. Given all the above, ALISE is compared with the following concurrent works: U-TAE[14] [3] (FS), Presto[15] [9] (FT and FR), U-BARN [15] (FS, FT, and FR), and U-BARN-GF (FR). This last approach is a variant of U-BARN, providing, as ALISE, fixed-dimensional SITS representations.

[13] https://pytorch.org/docs/stable/generated/torch.nn.CrossEntropyLoss.html

[14] https://github.com/VSainteuf/utae-paps

[15] https://github.com/nasaharvest/presto (commit 5486fd5)

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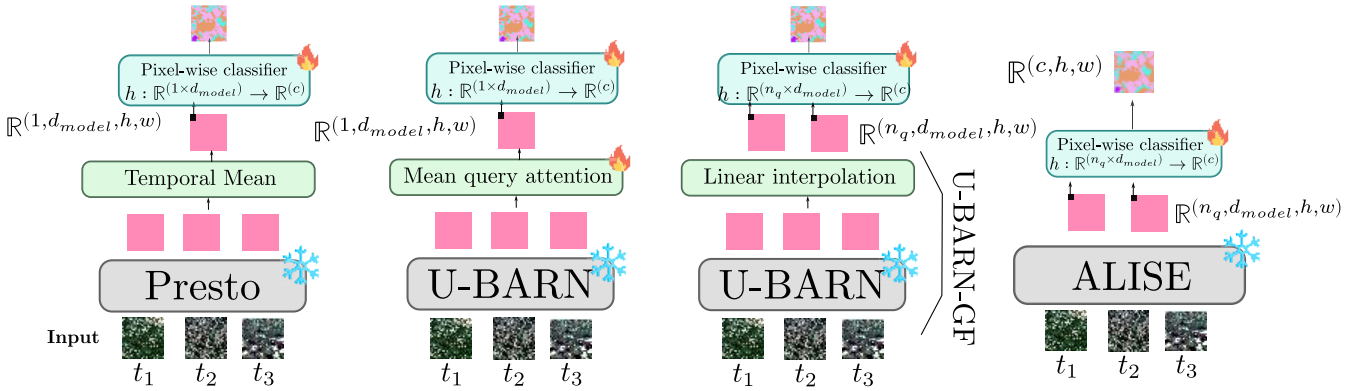


Fig. 8. Comparison of Presto, ALISE, U-BARN, and U-BARN-GF when frozen for the semantic segmentation task.

1) U-TAE is an FS architecture composed of a Unet network with a lightweight temporal attention mechanism located at the bottleneck.
2) Presto is a lightweight temporal SITS encoder pretrained as an MAE. It takes as input a monthly synthesis. The authors suggest selecting the least cloudy scene of each month. In contrast, as usually operated in RS, we train Presto with SITS composed of the median value of each band among the cloud-free acquisitions of each month. As suggested in [9], to exploit the latent representations provided by Presto, a temporal mean is performed.
3) U-BARN [15] is a spatio-spectro-temporal SITS encoder pretrained as an MAE. As U-BARN does not encode SITS into a fixed-size latent representation, the shallow classifier with a mean query attention mechanism proposed in [15] is considered here. Compared to the original implementation, we have modified the PE so that U-BARN can process multiyear SITS. Besides, we have pretrained U-BARN on MMDC-EU with the same pretraining configuration as ALISE. We call these SSL models U-BARN^{FT} and U-BARN^{FR} to denote the fine-tuning and frozen configurations.
4) *U-BARN-GF:* To assess the effectiveness of the proposed learnable temporal projector, we compare ALISE with an encoder composed of U-BARN followed by a linear interpolation layer, denoted U-BARN-GF. The irregular and unaligned representations from U-BARN are projected into $n_q = 10$ regularly spaced reference dates in the temporal extent of the downstream task. The resulting aligned representations are then processed through a single fully connected layer as done with ALISE. We denote U-BARN-GF^{FR} the configuration where U-BARN is pretrained and frozen during the downstream task. If the learnable projector is effective, we expect ALISE^{FR} to outperform U-BARN-GF^{FR}.

2) Change Detection Baseline: As indicated in Fig. 7, we propose to exploit ALISE representations without additional learning steps to perform change detection. To establish a fair comparison, ALISE is also compared to an unsupervised change detection strategy. In the proposed competitive work, the input SITSs are interpolated onto a fixed annual common time grid using a linear interpolation (gap-filling) method. Specifically, we interpolate the SITS valid acquisitions onto a regular temporal grid with a period of five days. The distance map is computed between the interpolated raw SITSs.

V. Experiments

This section evaluates the representations provided by the pretrained ALISE on three downstream tasks and compares them to competitive methods. First, we present a detailed analysis of ALISE's performance in frozen configurations for the two segmentation datasets (PASTIS and MultiSenGE). We also examine the effectiveness of the pretraining under a severe label scarcity scenario. Additionally, ALISE representations are assessed on an unsupervised change detection task with the CropRot dataset. Next, we provide an extensive discussion on the influence of several pretraining parameters (t_w, n_q, w_{rec}, w_{inv}, and w_{cov}). Finally, we propose a qualitative assessment of the role of the learnable queries in the temporal projector.

A. Segmentation Task Results

Two different segmentation tasks are employed to compare the SITS representations from the pretrained and frozen ALISE model with competitive works. In order to bring some perspective to the results obtained, the FS and FT configuration results are also shown. The two downstream datasets differ on two main points. First, in the MultiSenGE dataset, semantic labeling is dense, whereas in PASTIS, all pixels not belonging to a known crop are not classified. Consequently, we assume that spatial context must be better taken into account to succeed in the MultiSenGE task than in PASTIS. However, we assume that to distinguish between the 18 PASTIS crop classes compared to the 14 land cover classes of MultiSenGE, more complex temporal features are required. Table IV presents the averaged $F1$ score, the overall accuracy (OA), and the mean intersection over union (mIoU) on the PASTIS and MultiSenGE segmentation datasets. For each segmentation task, detailed $F1$ scores per class are displayed in Tables V and VI. To better understand the performance differences between ALISE and Presto, the confusion matrix is shown in Fig. 9.

First, although this article does not focus on the construction of a novel FS framework for SITS, ALISE^{FS} architecture

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TABLE IV

$F1$, MIOU AVERAGED PER CLASS, AND OA ON PASTIS AND MULTISENGE DATASETS. THE MEAN OF THE METRICS IS OBTAINED ON PASTIS' FIVEFOLD. ON THE MULTISENGE DATASET, TWO TRAININGS ARE CONDUCTED WITH DIFFERENT SEEDS. EACH COLOR CORRESPONDS TO A PRETRAINING CONFIGURATION, AND THE HIGHEST SCORE WITHIN A CONFIGURATION IS UNDERLINED. AS NO CLOUD MASKS ARE PROVIDED ON MULTISENGE, PRESTO CANNOT BE ASSESSED ON THIS SEGMENTATION TASK. THE NUMBER OF TRAINABLE PARAMETERS IS ESTIMATED ON THE PASTIS TASK

Name	Pre-training Data-Set	Train-able parame-ters	PASTIS F1	PASTIS OA	PASTIS mIoU	MSenGE F1	MSenGE OA	MSenGE mIoU
ALISE^{FR}	MMDC-EU	12.2K	0.68 ± 0.02	0.85 ± 0.01	0.56 ± 0.02	0.17 ± 0.00	0.57 ± 0.00	0.11 ± 0.00
ALISE^{FT}	MMDC-EU	1.1M	**0.81** ± 0.02	**0.91** ± 0.01	**0.70** ± 0.02	**0.23** ± 0.00	**0.63** ± 0.00	**0.16** ± 0.00
ALISE^{FS}	✗	1.1M	0.80 ± 0.01	**0.91** ± 0.01	0.69 ± 0.01	0.21 ± 0.00	0.62 ± 0.01	0.15 ± 0.00
Presto^{FR}	worlwide	2.5K	0.27 ± 0.01	0.62 ± 0.00	0.19 ± 0.01	✗	✗	✗
Presto^{FT}	worlwide	404K	0.55 ± 0.02	0.76 ± 0.01	0.41 ± 0.02	✗	✗	✗
U-BARN-GF^{FR}	MMDC-EU	12.2K	0.60 ± 0.01	0.82 ± 0.01	0.48 ± 0.01	0.12 ± 0.00	0.55 ± 0.00	0.09 ± 0.00
U-BARN^{FR}	MMDC-EU	13.8K	0.59 ± 0.03	0.82 ± 0.01	0.47 ± 0.03	0.14 ± 0.00	0.56 ± 0.00	0.10 ± 0.00
U-BARN^{FT}	MMDC-EU	1.1M	**0.81** ± 0.02	**0.91** ± 0.01	**0.70** ± 0.02	**0.23** ± 0.01	0.62 ± 0.00	**0.16** ± 0.01
U-BARN^{FS}	✗	1.1M	0.80 ± 0.01	0.90 ± 0.01	0.69 ± 0.02	**0.23** ± 0.01	0.62 ± 0.00	**0.16** ± 0.01
U-TAE	✗	1.1M	**0.81** ± 0.02	**0.91** ± 0.01	**0.70** ± 0.03	0.15 ± 0.02	0.61 ± 0.01	0.11 ± 0.01

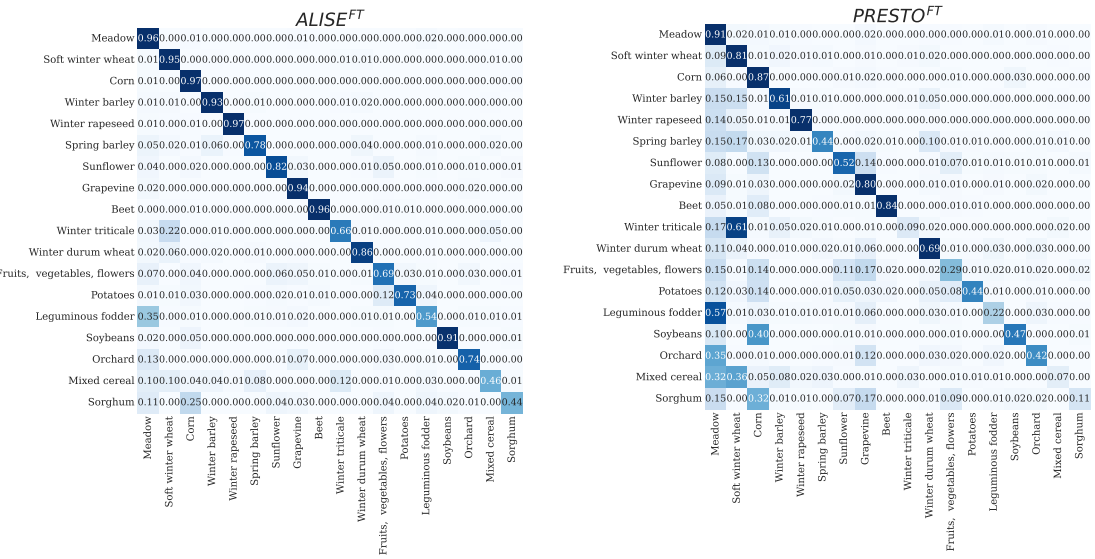


Fig. 9. Confusion matrices on the PASTIS crop segmentation dataset obtained after fine-tuning. For each confusion matrix, rows correspond to true labels and columns to predictions. The matrices are normalized per row. (Left) ALISE^{FT}. (Right) Presto^{FT}.

achieves performances consistent with current SOTA (U-TAE and U-BARN^{FS}). Next, Table IV demonstrates that ALISE^{FR} outperforms the existing models such as Presto^{FR} and U-BARN^{FR} on the PASTIS dataset. Compared with U-BARN, we observe a reduced gap between the frozen network and the FS or FT versions. Additionally, the frozen ALISE configuration drastically decreases the number of trainable weights from 1.1 M to 12.2k, highlighting that higher pretraining quality results in a better tradeoff between trainable weights and performance. Besides, compared to the PASTIS segmentation task, performances are lower on MultiSenGE. This may be explained by the fact that this last dataset is highly imbalanced, and minority classes decrease the macro-averaged mIoU and $F1$ scores, as shown in Table VI. Furthermore, the FT configuration (ALISE^{FT}) does not significantly outperform the FS approach (ALISE^{FS}). This finding is nonetheless consistent with the previous study [15] conducted on U-BARN. Differences between ALISE and the other two competitive works are illustrated in Fig. 8 and further detailed as follows.

1) ALISE Versus U-BARN: Segmentation metrics detailed in Table IV show that ALISE^{FR} outperforms U-BARN^{FR}: (+9% $F1$), (+3% 0A), and (+9% mIoU) on the PASTIS dataset and (+3% $F1$), (+1%OA), and (+1% on mIoU) on MultiSenGE. Remarkably, ALISE significantly outperforms U-BARN in linear probing while having a shallower classifier (no learnable mean query) and smaller latent representations (ten latent temporal features). We also observe in Table V that the boost of performance may vary depending on the PASTIS crop classes. Several classes such as winter tricitale (+25%), sunflower (+14%), sorghum (+19%), and mixed cereal (+15%) exhibit stronger boost of performances compared to the other classes. The overall gain of performance can be explained by the differences between ALISE and U-BARN. ALISE differs from U-BARN in two main aspects: 1) its

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TABLE V

$F1$ SCORE PER CLASS ON PASTIS DATASET FOR EACH TRAINING CONFIGURATION. MEAN AND STANDARD DEVIATION OF THE $F1$ SCORE OBTAINED USING k-FOLD TRAINING ARE DETAILED

	ALISE^{FR}	**ALISE^{FT}**	**ALISE^{FS}**	**Presto^{FR}**	**Presto^{FT}**	**U-BARN^{FR}**	**U-BARN^{FT}**	**U-BARN^{FS}**	**U-TAE**
Meadow	0.91 ± 0.01	**0.94** ± 0.01	**0.94** ± 0.01	0.75 ± 0.01	0.85 ± 0.01	0.90 ± 0.01	**0.94** ± 0.01	**0.94** ± 0.01	**0.94** ± 0.01
Soft winter wheat	0.89 ± 0.01	**0.94** ± 0.01	**0.94** ± 0.01	0.65 ± 0.03	0.79 ± 0.02	0.86 ± 0.01	**0.94** ± 0.01	**0.94** ± 0.01	**0.94** ± 0.01
Corn	0.93 ± 0.01	**0.96** ± 0.01	**0.96** ± 0.00	0.71 ± 0.01	0.85 ± 0.01	0.91 ± 0.01	**0.96** ± 0.01	**0.96** ± 0.01	0.96 ± 0.01
Winter barley	0.82 ± 0.02	**0.92** ± 0.01	**0.92** ± 0.02	0.21 ± 0.04	0.67 ± 0.03	0.77 ± 0.04	**0.92** ± 0.02	**0.92** ± 0.02	**0.92** ± 0.01
Winter rapeseed	0.91 ± 0.01	**0.96** ± 0.01	**0.96** ± 0.01	0.49 ± 0.04	0.81 ± 0.02	0.87 ± 0.04	**0.96** ± 0.01	**0.96** ± 0.01	**0.96** ± 0.01
Spring barley	0.67 ± 0.08	**0.79** ± 0.05	0.77 ± 0.06	0.01 ± 0.01	0.49 ± 0.07	0.61 ± 0.09	0.78 ± 0.06	0.76 ± 0.06	0.77 ± 0.05
Sunflower	0.71 ± 0.03	**0.83** ± 0.01	**0.83** ± 0.02	0.18 ± 0.03	0.56 ± 0.05	0.57 ± 0.07	**0.83** ± 0.04	0.81 ± 0.03	**0.83** ± 0.05
Grapevine	0.82 ± 0.02	**0.92** ± 0.01	0.91 ± 0.01	0.57 ± 0.06	0.71 ± 0.05	0.77 ± 0.03	0.91 ± 0.01	0.91 ± 0.00	0.91 ± 0.01
Beet	0.93 ± 0.02	**0.96** ± 0.01	0.95 ± 0.02	0.45 ± 0.10	0.85 ± 0.02	0.89 ± 0.01	**0.96** ± 0.01	**0.96** ± 0.02	**0.96** ± 0.01
Winter triticale	0.41 ± 0.06	0.70 ± 0.06	0.66 ± 0.06	0.00 ± 0.00	0.15 ± 0.05	0.16 ± 0.06	0.69 ± 0.04	0.65 ± 0.07	**0.73** ± 0.04
Winter durum wheat	0.74 ± 0.03	**0.83** ± 0.03	0.81 ± 0.03	0.41 ± 0.02	0.64 ± 0.03	0.69 ± 0.01	0.82 ± 0.03	0.80 ± 0.02	0.82 ± 0.04
Fruits/veg/flowers	0.49 ± 0.07	0.70 ± 0.03	0.65 ± 0.03	0.04 ± 0.02	0.34 ± 0.10	0.35 ± 0.05	**0.71** ± 0.03	0.65 ± 0.03	0.69 ± 0.06
Potatoes	0.65 ± 0.07	**0.76** ± 0.08	0.70 ± 0.05	0.10 ± 0.06	0.52 ± 0.09	0.55 ± 0.09	0.74 ± 0.03	0.70 ± 0.06	0.71 ± 0.09
Leguminous fodder	0.44 ± 0.08	0.60 ± 0.07	**0.64** ± 0.07	0.10 ± 0.02	0.30 ± 0.05	0.35 ± 0.11	0.63 ± 0.07	0.63 ± 0.09	0.61 ± 0.08
Soybeans	0.82 ± 0.04	0.92 ± 0.02	**0.93** ± 0.02	0.01 ± 0.02	0.54 ± 0.06	0.71 ± 0.07	**0.93** ± 0.02	0.92 ± 0.02	0.92 ± 0.03
Orchard	0.61 ± 0.05	0.77 ± 0.04	0.77 ± 0.04	0.12 ± 0.04	0.47 ± 0.07	0.55 ± 0.04	0.77 ± 0.05	0.77 ± 0.04	**0.79** ± 0.04
Mixed cereal	0.23 ± 0.08	0.53 ± 0.07	0.53 ± 0.05	0.00 ± 0.00	0.11 ± 0.04	0.08 ± 0.05	0.54 ± 0.07	0.54 ± 0.05	**0.55** ± 0.06
Sorghum	0.29 ± 0.08	0.50 ± 0.11	0.50 ± 0.09	0.00 ± 0.00	0.18 ± 0.08	0.10 ± 0.09	0.52 ± 0.11	0.49 ± 0.12	**0.53** ± 0.13

encoder provides fixed-size aligned representations and 2) the pretraining strategy is different. As detailed in Section III-A, ALISE corresponds to the U-BARN architecture on top of which is placed a temporal projector. Experiments detailed in Section V-D show that ALISE's pretraining is primarily driven by its cross-reconstruction task, which is close to U-BARN's MAE pretraining. Therefore, we believe that the improvement in performance when freezing the pretrained SITS encoder is largely due to the inclusion of the temporal projector in ALISE. Furthermore, to ensure that these improved results are not caused by the use of the shallow classifier architecture combined with ALISE, ALISE^{FR} performances are compared with U-BARN-GF. Similar to ALISE, U-BARN-GF aligned features are injected into a fully connected layer that operates on the spectro-spatio-temporal features. We observe in Table IV that U-BARN^{FR} and U-BARN-GF^{FR} have close performances and that ALISE^{FR} outperforms U-BARN-GF^{FR} on both datasets. This result demonstrates the effectiveness of using a learnable temporal projector over a linear interpolation strategy. Furthermore, by design, ALISE provides more relevant representations in the frozen configuration than U-BARN.

2) ALISE Versus Presto: We observe that ALISE^{FR} outperforms both frozen and FT Presto configurations, by 41.5% and 13.6%, respectively. To study more deeply the obtained performances, we compare the confusion matrices obtained by ALISE^{FT} and Presto^{FT} in Fig. 9. We observe that ALISE and Presto show similar confusions between classes: winter triticale and soft winter wheat, leguminous fodder and meadow, and corn and sorghum. Nevertheless, the confusion values between classes obtained by ALISE are each time lower than values reached by Presto. We posit that this result can be explained by several factors. First, Presto is a lightweight spectrotemporal architecture that does not take spatial context into account. Such a design may not be relevant to segmentation tasks. Additionally, due to the required undersampling protocol (Presto exploits monthly synthesis instead of all available acquisitions), it may miss key temporal information in comparison to ALISE.

B. Label Scarcity Scenario

To assess the behavior of the model in a severe label scarcity scenario, a reduced version of the PASTIS dataset has been created. Following the approach of [15], we have used five smaller datasets, each composed of 30 SITS for each PASTIS fold. Therefore, the results shown in Table VII correspond to the averaged macro $F1$ score across 25 trials. Under severe label scarcity, the FT model outperforms the FS framework by 12.5%. Interestingly, the frozen ALISE also outperforms its FS configuration by 9.7%. Given its reduced number of pretrainable parameters compared to FS and FT approaches, ALISE^{FR} is an ideal candidate for scenarios with limited labeled data.

C. Change Detection Task

To quantitatively compare the change detection performances on the proposed *CropRot* dataset, we compute the area under the receiver operating characteristic curve (AUC) score. This score is calculated on the distance map computed between the representations of SITS from two different years. Table VIII shows that better change detection results are obtained by comparing SITS representations encoded by ALISE. Besides, in contrast to the linear interpolation on the raw input SITS, the ALISE change detection framework does not require cloud mask information. Furthermore, our results demonstrate that ALISE's representations are relevant for change detection, even though its PE information is absolute and not relative to a year (DOY). In other words, ALISE can still learn SITS invariance between different years, even if the PE differs for each of the two years being compared. In addition, a qualitative analysis of the change detection maps is performed. Given two annual irregular and unaligned SITSs from 2019 and 2020, Fig. 10 illustrates the obtained change maps. In this example, SITS, the number of available spring acquisitions is greater in 2020 than in 2019. Besides, Fig. 10 shows class variability even when there is no change. These variations can be caused by different agricultural practices, meteorological events, and different acquisition dates. In Fig. 10, this intraclass variability is observed when looking

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TABLE VI

$F1$ SCORE PER CLASS ON THE MULTISENGE SEGMENTATION TASK FOR EACH CONFIGURATION. TWO TRAININGS ARE PERFORMED FOR EACH CONFIGURATION

	ALISE^{FS}	**ALISE^{FT}**	**ALISE^{FR}**	**U-BARN^{FR}**	**U-BARN^{FS}**	**U-BARN^{FT}**	**U-TAE**
Dense Built-Up	**0.04** ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Sparse Built-Up	0.11 ± 0.02	0.13 ± 0.00	0.02 ± 0.01	0.01 ± 0.00	0.14 ± 0.02	0.14 ± 0.02	**0.15** ± 0.01
Specialized Built-Up Areas	0.13 ± 0.00	**0.31** ± 0.01	0.07 ± 0.00	0.02 ± 0.00	0.27 ± 0.03	0.27 ± 0.09	0.02 ± 0.02
Specialized but Vegetative Areas	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Large Scale Networks	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Arable Lands	0.67 ± 0.01	**0.68** ± 0.00	0.62 ± 0.00	0.62 ± 0.00	**0.68** ± 0.00	**0.68** ± 0.00	0.67 ± 0.01
Vineyards	0.46 ± 0.04	**0.51** ± 0.01	0.32 ± 0.00	0.21 ± 0.04	0.47 ± 0.04	0.48 ± 0.02	0.19 ± 0.27
Orchards	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Grasslands	0.39 ± 0.04	0.39 ± 0.00	0.33 ± 0.01	0.31 ± 0.02	0.40 ± 0.01	**0.41** ± 0.01	0.40 ± 0.00
Groces,Hegdes	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Forest	0.72 ± 0.01	**0.73** ± 0.00	0.67 ± 0.00	0.66 ± 0.00	**0.73** ± 0.00	**0.73** ± 0.00	0.71 ± 0.02
Open Spaces,Mineral	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Wetlands	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

TABLE VII

MACRO-AVERAGED $F1$ SCORES OBTAINED ON PASTIS ON LABELED DATA SCARCITY SCENARIO. EACH PASTIS FOLD IS COMPOSED OF 30 LABELED SITS

Model	F1
ALISE^{FR}	0.44 ± 0.01
ALISE^{FT}	**0.47** ± 0.04
ALISE^{FS}	0.34 ± 0.06

TABLE VIII

AUC METRIC ON CROPROT

Input Data	AUC
ALISE representations	**0.91**
raw interpolated SITS	0.88

at the fields located at the center bottom of the SITS (red circle). Although the crop class of this field has not changed between 2019 and 2020, we can visually observe important differences between May 14, 2019 and May 18, 2020, which are supposed to be close acquisitions. Nevertheless, the distance map shown in Fig. 10 is not affected by such input intraclass variability. In addition, compared to the distance map obtained from interpolated raw SITS, the distance on ALISE representations better distinguishes modified crops from unchanged ones.

D. Co-Influence of t_w, w_{inv}, w_{cov}, and w_{rec}

t_w is a hyperparameter involved in the view generation process and detailed in Section III-B1. More precisely, t_w is the number of acquisitions contained in the temporal window used to build each view. Increasing t_w is supposed to create greater discrepancies between views, therefore impacting both the discriminative and the cross-reconstruction losses. As t_w is the number of consecutive acquisitions, when t_w is increased, the cross-reconstruction task is no longer a simple interpolation task. Therefore, we aim to assess the co-influence of the view generation protocol (controlled by t_w) and the loss weights. Different pretraining configurations evaluating the impact of the four parameters (t_w, w_{inv}, w_{cov}, and w_{rec}) have been performed. For each configuration, results obtained on the five PASTIS folds are averaged by considering four different pretrained models with different seeds. Only the loss weights and the t_w parameter are evaluated; all the rest of hyperparameters are fixed for the rest of pretrained model configurations. The covariance weight value is set to 0.05 to reproduce the balance between the invariance and covariance losses suggested in VicReg [45]. The influence of t_w is analyzed by studying the macro-averaged $F1$ score before providing a more precise analysis per crop class.

1) Macro-Averaged F1 Score: Fig. 11 displays the linear-probing performances as a function of t_w. This figure shows that the additional invariance latent loss (L_{inv}) significantly degrades the linear-probing performances for t_w greater than 2 (the orange and blue curves are lower than the green one). We assume that the invariance loss might constrain too much the total loss too much when pretraining data are composed of large dissimilar views. For $t_w = 2$, there seems to be a slight improvement in the linear-probing performances when invariance loss is incorporated. Finally, these experiments do not show any benefit from using the covariance loss (L_{cov}) in addition to the invariance loss. There are several possible explanations for this outcome. First, the large memory size of SITS limits the batch size; therefore, experiments have been conducted with a batch size equal to 2. Although we use $b \times n_q \times h \times w$ samples to estimate the covariance matrix, these samples are correlated. In the original VicReg implementation [45], the covariance was estimated across 2048 samples, each corresponding to a different image. Second, the covariance loss in VicReg plays a crucial role in preventing information collapse. In our framework, the cross-reconstruction loss prevents collapsing, making the covariance loss less important during pretraining. Third, the projector architecture may impact the computation of the covariance loss. Therefore, more experiments studying the impact of the batch size and the projector could be necessary. Finally, the green curve in Fig. 11 depicts the influence of t_w when solely the reconstruction loss is applied. In this case, the downstream segmentation performance is also impacted by t_w. With large temporal windows ($t_w = 10$), the pretraining reconstruction task may become too complex, which prevents the model from learning informative SITS representations. Surprisingly, with smaller values of $t_w \leq 5$, no major differences are observed.

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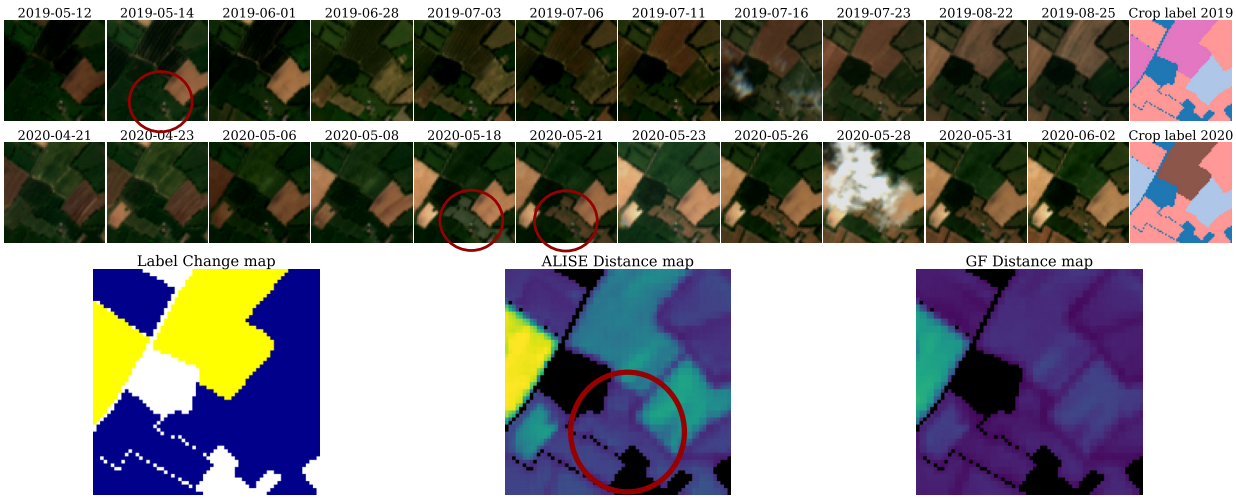


Fig. 10. Visualization of a distance map obtained on the change detection dataset with the pretrained ALISE. The top and second rows represent a portion of the S2 SITS along with their crop classes for 2019 and 2020, respectively. These represented images have similar index positions within their SITS. In the crop label maps, dark blue represents the background class. On the bottom row, the distance maps computed from the aligned representations from ALISE and the gap-filling methods are shown. The same scale is used in the colorbar of the distance maps. Pixels that belong to the background class are masked. On the bottom row, the leftmost image represents the ground truth change map with, in white the background class, in blue the *no change* label, and in yellow the *change* label.

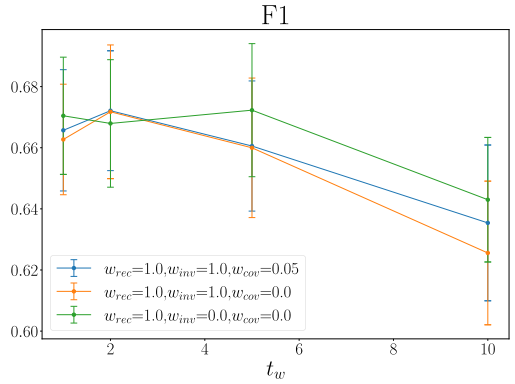


Fig. 11. Segmentation task performances on PASTIS linear probing as a function of t_w. In all these experiments, $n_q = 10$, four pretrainings were conducted and their performances on five of PASTIS folds were evaluated.

This could be explained by the fact that, unlike regular time series processing, t_w does not control the temporal extent that is reconstructed. Inherent important temporal gaps in S2 SITS might provide a complex pretraining task even with $t_w = 1$.

2) F1 Score per Class: We propose a more in-depth analysis of the effect of t_w and the pretraining loss weights in Fig. 12. Notably, similar to the previous experiment, the $F1$ score for each PASTIS crop class is plotted as a function of t_w. Different behaviors are observed depending on the crop classes. For many crop classes, there is a decrease in the $F1$ score when t_w increases. However, some crop classes such as meadow, corn, spring barley, grapevine, fruits, vegetables and flowers, potatoes, leguminous fodder, and orchard are unaffected by t_w. With the exception of grassland, maize, and spring barley, we hypothesize that the lack of t_w effect for these classes may be linked to the fact that they are either permanent (grapevine and orchards) or greenhouse. Interestingly, the soybean class exhibits an outlier behavior, with an increase in $F1$ score as t_w increases. This experiment demonstrates that the influence of pretraining conditions differs depending on the target class.

E. Impact of n_q

For practical purposes, it is relevant to reduce the size of the latent representation (n_q) while preserving the downstream tasks' performances. Fig. 13 plots the segmentation performances on the PASTIS dataset as a function of n_q. For each configuration, one pretraining was conducted, and the performances were assessed on one out of the five available PASTIS experiments. We observe that increasing the value of n_q improves the downstream task $F1$ score. This can be attributed to two factors. First, a greater value of n_q means a larger latent space, and therefore, potentially more information contained within it. Second, it is assumed that a smaller value of n_q makes the cross-reconstruction task more difficult due to the compression performed in the proposed temporal projector. This compression could penalize the cross-reconstruction pretraining task. Nevertheless, these comparisons with different possible values of n_q may not be totally fair. Indeed, a higher value of n_q results in a larger classifier during linear probing, which may have a positive impact on downstream task performances.

F. Qualitative Analysis of the Temporal Projector

To gain a better understanding of the information encoded by ALISE, we conduct a qualitative analysis of the latent representations. For this purpose, we propose to study how latent information is used by the self-attention mechanism of the decoder during the reconstruction process. We note the pixel-level latent temporal vector, indexed by i, as $\mathbf{y}_{(\mathbf{i},.)}$. To understand the importance of each latent temporal feature during the reconstruction process, the attention weights of each decoder head are displayed in Fig. 14. For improved

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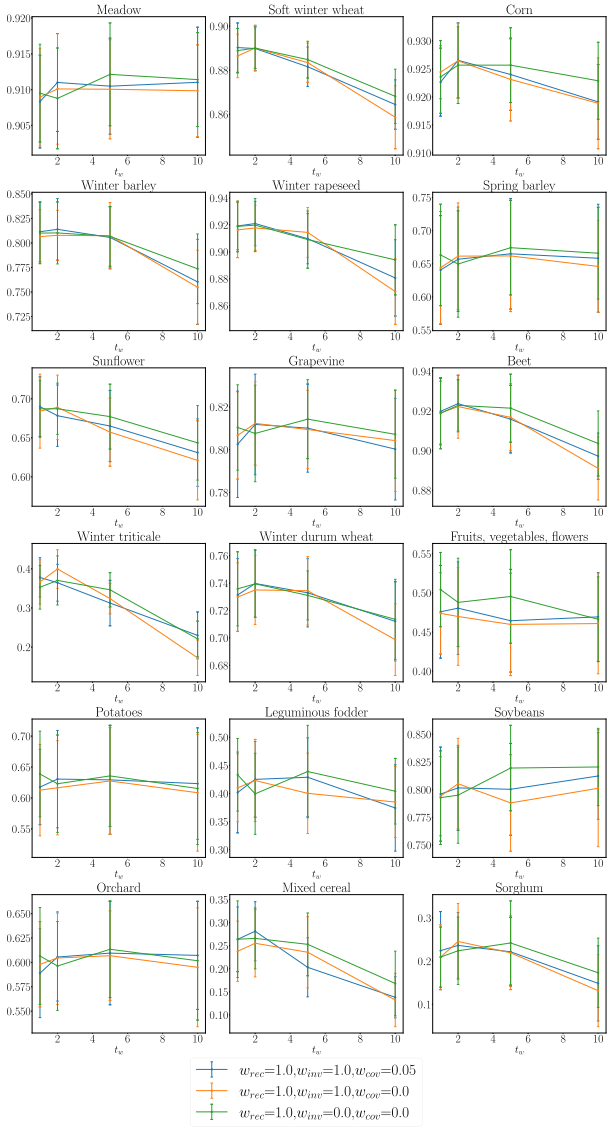


Fig. 12. $F1$ score per class on PASTIS linear probing as a function of t_w. In all these experiments, $n_q = 10$. Results are averaged over four pretrained models for all five PASTIS folds.

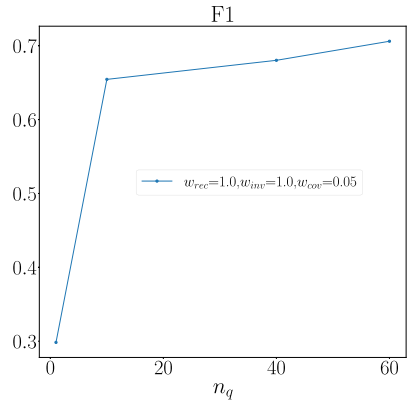


Fig. 13. Segmentation task performances in linear-probing configuration on the PASTIS dataset as a function of n_q. In these experiments, $t_w = 5$. For each configuration, one pretraining is conducted and the downstream task is performed on one out of five PASTIS experiments.

visualization, the reconstruction decoder is asked to reconstruct in Fig. 14, a regularly sampled time series from January 1, 2017 to July 19, 2021 with a step of ten days. The reconstruction temporal grid considered here corresponds to the years observed during pretraining (2017–2020) as well as years outside the temporal extent of the pretraining dataset (2021). In an attention matrix, a high attention score at a given row (indexed by i) and column (corresponding to a date d_j) indicates the importance of latent temporal feature $\mathbf{y}_{(\mathbf{i},.)}$ for the reconstruction of the date d_j. For each latent temporal feature (row), high attention scores (bright color) are often observed on specific narrow intervals, while attention weights are low outside of them. A notable finding is that for a latent temporal feature, these intervals are often separated by approximately one year. Although no annual periodicity is explicitly given as input to the decoder, dates to reconstruct spaced by 365 days exhibit a similar high attention score on the same latent temporal feature. Furthermore, it is worth noting that these annual periodic patterns are also observed for reconstructed dates not included in the pretraining (year 2021), suggesting that the model might have forecasting (extrapolation) abilities.

The influence of the latent temporal features on the reconstruction is also illustrated in Fig. 15. Specifically, the reconstruction is conducted with either all latent temporal features, a single latent temporal feature, or a triplet of features. The top row depicts random acquisition dates within the input SITS, while the next rows display the reconstruction of the decoder of the SITS on dates ranging from January 1, 2017 to November 11, 2021, with acquisitions spaced by 70-day intervals. Consequently, the image acquisitions of the first row are not temporally aligned with the rows below. Nevertheless, the predictions generated by the model using all latent temporal features (second row) and the input SITS (first row) are often consistent. For example, the acquisitions marked by the blue, green, and orange rectangles are temporally close to some of the input predictions. These latter acquisitions show coherent reconstructions, which highlights that the use of the temporal alignment projector does not lead to a significant loss of information. The following three rows in Fig. 15 illustrate reconstruction results obtained when a single latent temporal feature is used during the decoding stage, $\mathbf{y}_{(\mathbf{9},.)}$, $\mathbf{y}_{(\mathbf{8},.)}$, and $\mathbf{y}_{(\mathbf{5},.)}$. For each row, it can be observed that the model invariably generates the same image, revealing that the temporal dynamics are not reconstructed. Conversely, the temporal dynamic can be observed in the reconstructions obtained from only three latent temporal features (last row). For this case, the contribution of each latent temporal feature to the image reconstruction process is easily recognizable. In the proposed example, the reconstruction obtained from $\mathbf{y}_{(\mathbf{9},.)}$ highly contributes to the reconstruction of winter images shown on the last row. The latent temporal features $\mathbf{y}_{(\mathbf{8},.)}$ and $\mathbf{y}_{(\mathbf{5},.)}$ seem to intervene for May–July and July–November, respectively. In this last row, the majority of the images appear to be highly similar to the reconstruction obtained from one single latent temporal feature. Nevertheless, a few acquisitions seem to be obtained through the mix of information from different latent temporal features. For instance for May 21, 2017, it seems that the latent temporal features $\mathbf{y}_{(\mathbf{8},.)}$ and $\mathbf{y}_{(\mathbf{9},.)}$ are merged. Areas showing this combination are marked by

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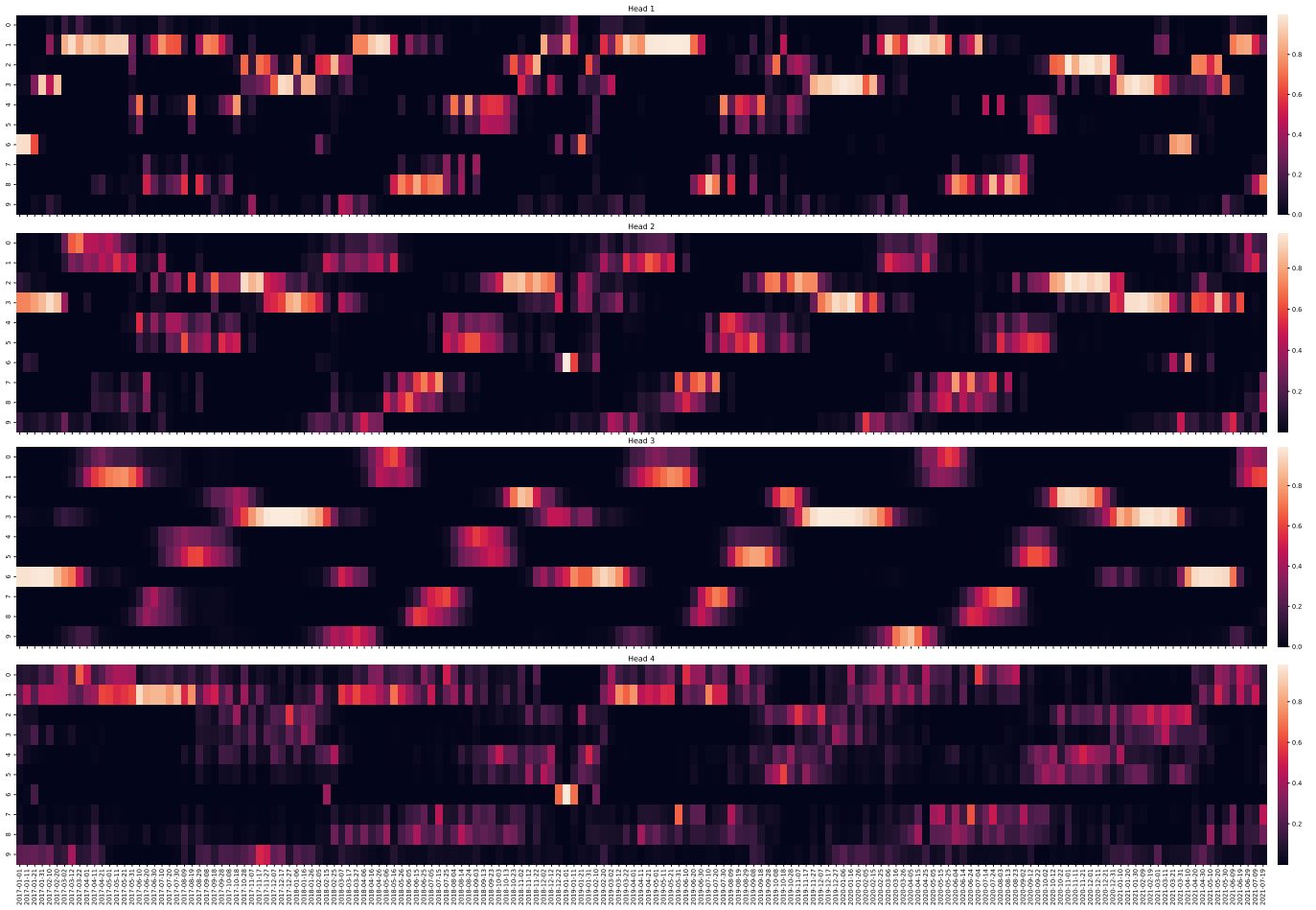


Fig. 14. Attention weights of the reconstruction decoder when reconstructing a time series from January 1, 2017 to July 19, 2021. Attention matrices plotted correspond to the average attention matrix obtained for each pixel of the SITS. Each matrix from top to bottom corresponds to a different head. On each score matrix, the column corresponds to the latent temporal feature, while the row corresponds to the dates to reconstruct.

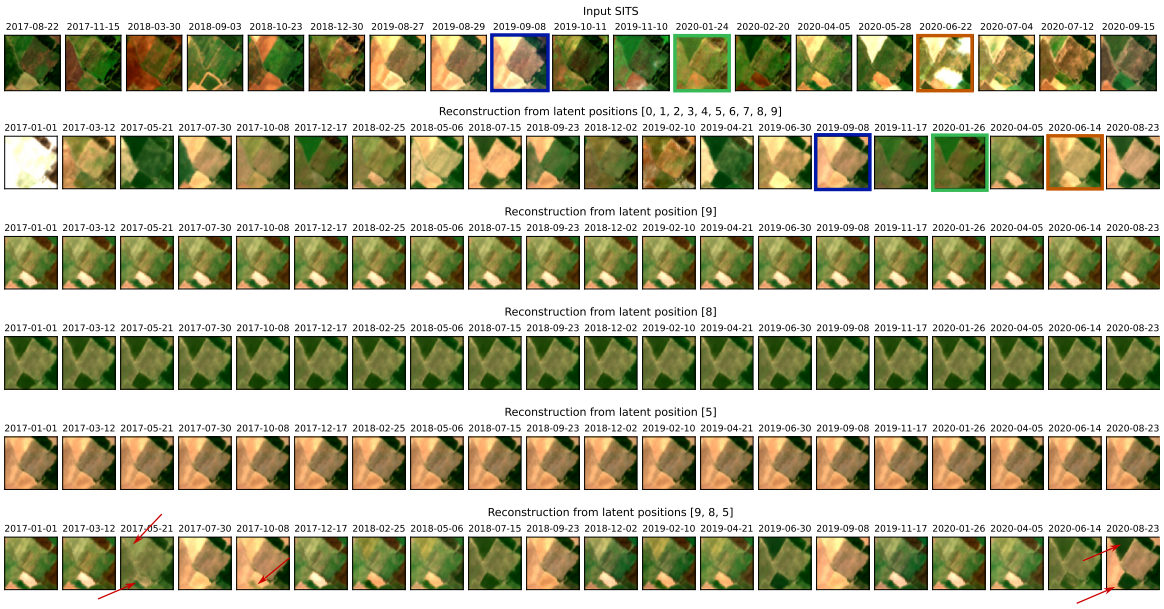


Fig. 15. Reconstructions obtained from the decoder given different latent temporal features configurations. (First row) Random dates from the input SITS. Then, reconstruction obtained by using: all latent temporal features (second row), latent temporal feature no. 9 (third row), latent temporal feature n°8 (fourth row), latent temporal feature no. 5 (fifth row), and latent temporal features 9, 8, and 5 (bottom row).

the red arrows in Fig. 15. From these observations, we may consider the aligned representations as a novel basis for input SITS representation. From this point of view, latent temporal feature data would serve as a prototype and the attention

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weights in the decoder would fulfill the role of membership degree.

VI. Conclusion

This article discusses the notable challenges involved in learning to represent SITS. In particular, our work paves the way for the construction of an FM for land surface monitoring using Earth observation.

This article proposes a new SITS encoder named ALISE, which exploits spatial, spectral, and temporal dimensions and generates aligned and fixed-size representations of irregular and unaligned multiyear SITS. ALISE is pretrained using a new multiview hybrid SSL pretraining task that combines MAE loss with instance discrimination losses. In addition, the ALISE pretraining dataset (MMDC-EU) is a custom-built large-scale multiyear unlabeled dataset. The quality of ALISE's representation has been assessed on three downstream datasets: PASTIS (crop segmentation), MultiSenGE (dense land cover segmentation), and the novel CropRot (crop change detection). Our results demonstrate the significant progress made with regard to the three representation characteristics considered: *easy to use*, *informative*, and *generic*.

First, ALISE provides aligned, fixed-size representations that preserve the spatial resolution of the input data. Consequently, ALISE representations can be easily exploited by a shallow classifier. In this article, a single fully connected layer was used to perform two segmentation tasks (crop mapping and dense land cover). Results have also shown that pretrained and frozen ALISE outperforms the FS approach when labeled data are scarce, highlighting the value of our self-supervised pretraining. The remarkable performance of frozen and pretrained ALISE indicates that an important step has been taken toward the creation of *ready-to-use* SITS representations. The production of aligned, fixed-size representations has been achieved through the use of a learnable query-based cross-attention mechanism. We have also provided the first qualitative study of the aligned latent temporal features obtained through this latter mechanism. It appears that each latent temporal feature summarizes a specific part of the input SITS. To reconstruct an SITS, the pretraining decoder successfully recovers the annual periodicity of the SITS, whereas our temporal encoding does not rely on the DOY.

Second, the quality of ALISE representations was compared with existing competitive works. Results obtained by pretrained and frozen ALISE outperform Presto [9] and U-BARN [15] for both semantic segmentation tasks. As a result, ALISE's representations may be considered more *informative* than other existing works. In addition, we have studied in depth our proposed hybrid SSL approach. Notably, the impact of the view generation method and the contribution of each loss have also been investigated. Our results show that most of the pretraining is driven by the cross-reconstruction task. Nevertheless, depending on how the view generation is performed, which is strongly influenced by t_w, the invariance loss may or may not improve performance. This leads us to think that other view generation protocols could be investigated. Besides, our experiments did not reveal a significant contribution from the covariance loss. These unexpected findings also highlight the important challenges that remain in applying ideas from the wider computer vision field to the specificities of SITS (temporal dynamics, physics of the measure, and so on).

Third, the *genericity* of the representations was assessed on three proposed downstream tasks. In addition to the great performances obtained in both segmentation tasks, our results demonstrate that the proposed aligned SITS representations can be used for downstream unsupervised change detection tasks. We also consider the proposed novel crop change detection dataset named *CropRot*, as an important contribution to assess future FM on SITS. Besides, to learn **generic** representations, ALISE was pretrained on a new large-scale and multiyear dataset. Nevertheless, the geographical diversity of the pretraining and downstream datasets could be improved since pretraining and downstream data only include European geographical areas. The development of a scalable method, trained and evaluated on numerous geographical configurations, remains unexplored here.

It should also be noted, however, that this article does not address certain aspects. For instance, ALISE memory consumption is quadratic with the temporal size of the input SITS. Therefore, lightweight architectures based on learnable queries [62], [63], [64] could be considered. Additionally, although our proposed datasets represent a significant improvement over existing ones, building an FM for monitoring land surfaces requires additional ones. Consequently, there is an urgent need to develop pretraining datasets spanning larger geographical regions, as well as to create labeled datasets from diverse geographical areas. Without these efforts, there is a risk of developing biased FMs that are predominantly evaluated in wealthier northern countries, which tend to have more labeled data. Besides, in order to develop spatially scalable methods, integrating spatial encoding [65] or meteorological encoding information [66] could also be relevant. Finally, a major remaining challenge in developing FM is the processing of multisensor data. For example, combining S1 data with S2 data is beneficial when optical data are unavailable due to unsuitable weather conditions. Furthermore, the use of different modalities in a multiview SSL protocol is promising. It could provide natural SITS views and avoid a complex view generation protocol. As a result, a greater contribution from instance discrimination losses in this context may be observed.

Appendix

A. CropRot Additional Information

The change matrix between 2019 and 2020 is represented in Fig. 16. As expected, the rate of change depends on the considered class. We observe important rotations between cereal and corn, while grassland mostly remains unchanged.

B. Comparison With VicRegL

The proposed discriminative losses are similar to those of VicRegL [46]. However, three notable modifications have been introduced. First, unlike VicRegL, the invariance loss does not require any matching functions to realign the pixels of both views since geometric augmentation is not performed. In our case, each embedded vector at the pixel level is compared with

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Fig. 16. Change matrix between years 2019 and 2020 on the crop classes. Class correspondence is {5: rapeseed, 6: cereal, 9: sunflower, 10: corn, 11: rice, and 13: grassland}.

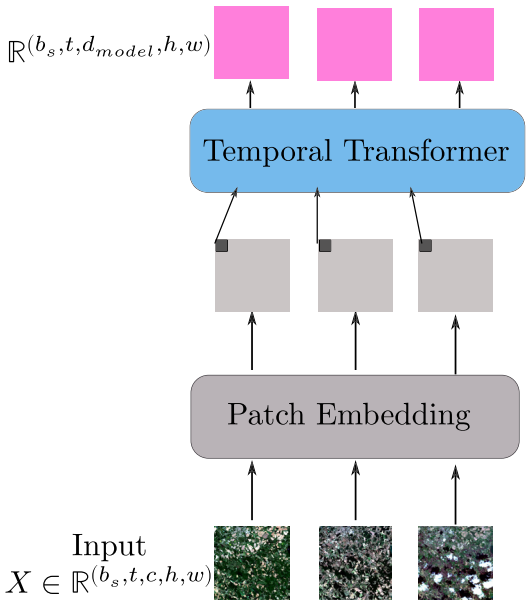


Fig. 17. U-BARN architecture, which is divided into two blocks. The "patch embedding" block is composed of a Unet network and the PE. The Unet processes independently each image of the SITS. Then, the temporal transformer processes pixel-level SITS.

the embedded vector of the other view at the same spatial position. Second, the large SITS size strongly constrains the batch size, which differs from the larger batch values of VicRegL. In VicRegL, the covariance loss is computed for each pixel of the latent representation using the b samples of the batch. The final local covariance loss is the sum over the spatial dimensions $h \times w$ of the pixel-level losses. Instead of estimating a covariance for each pixel, our covariance loss is estimated for the d_{emb} variables using $b \times h \times w$ samples. Third, the variance loss is not considered in our approach. If the variance was estimated by considering $b \times h \times w$ samples, keeping the variance of each variable above a threshold would enforce a strong variability between pixels that might come from the same image. This loss could then deteriorate the spatial consistency of the representation.

C. ALISE Architecture

1) U-BARN: Tables IX and X describe the architectural hyperparameters of the spatio-spectro-temporal encoder (Fig. 17).

TABLE IX

HYPERPARAMETERS OF THE ARCHITECTURE OF THE UNET ENCODER, WITH B AND T, RESPECTIVELY, THE BATCH AND TEMPORAL DIMENSIONS. THE *Down Block* ARCHITECTURE IS DETAILED IN [15]

Block Name	Input dimensions	Output dimensions
Input Convolution	(B*T,64,64,10)	(B*T,64,64,64)
Down Block 1	(B*T,64,64,64)	(B*T,32,32,64)
Down Block 2	(B*T,32,32,64)	(B*T,16,16,64)
Down Block 3	(B*T,16,16,64)	(B*T,8,8,128)

TABLE X

ARCHITECTURAL HYPERPARAMETERS OF THE TRANSFORMER IN U-BARN

N_{layers}	N_{head}	$attn_{dropout}$	dropout	d_{model}	d_{hidden}
3	4	0.1	0.1	64	128

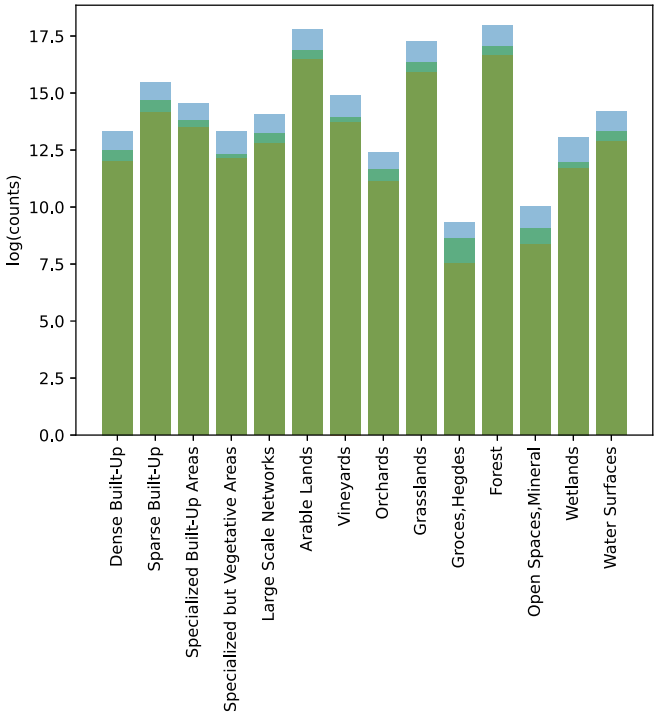


Fig. 18. Class distribution from MultiSenGE. Bar chart represents the log of the number of pixel belonging to each class.

2) Temporal Projector: The temporal projector is composed of a lightweight multihead cross-attention mechanism with two heads. Inspired by the attention mechanism proposed in [2], the channels of the input embeddings are distributed among the heads.

D. MultiSenGE Data Distribution

See Fig. 18.

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